

Visual Augmentation of Audio Semantics for Accessibility

Technical report

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Code and data: <https://github.com/adamgavriely/MscFinalProject>, release v1.1.0

Abstract

Subtitles carry speech, but a deaf or hard-of-hearing viewer still misses the other sounds of a video: a siren behind the camera, a dog barking in the next room. Captioning practice leaves out any sound whose source can be seen, so the sounds that matter most are the ones whose source is *off screen*. This project builds a system that finds those sounds and shows a generated picture of each one beside the video while it is heard.

Nothing is trained. The system chains ready-made open models: two sound detectors find candidate sounds, two audio-language models (“listeners”, asked for example “is a siren present?”) confirm weak ones, a vision-language model checks whether the source is visible on screen, and a text-to-image model draws the picture. Every rule that joins them was fixed on a development set before it was tested.

Scored per sound against human labels, the final system finds 24 of 65 needed off-screen sounds on a test set of 88 clips with 24 wrong pictures, a viewer cost of 2.41 per clip (a miss costs 4, a wrong picture 2). The baselines named in the project proposal do worse: generating a picture directly from the audio for every heard sound (or captioning it) costs 3.18, and showing nothing costs 2.96. The differences are significant (final system minus direct audio-to-image -0.77 per clip, 95 % interval $[-1.05, -0.50]$, $p < 0.001$; minus showing nothing -0.55 , $[-1.07, -0.02]$, $p = 0.041$). The gain comes from the on-screen check: it keeps most of the right pictures and removes most of the wrong ones.

Two limits remain. About one needed sound in five cannot be found even with perfect decisions, because the detectors barely hear it under speech or music; and the vision model confuses “the object is in the frame” with “the object is making this sound now”. The test set was read after each accepted change, so the test numbers are a report on a seen set, and no study with deaf viewers was run.

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1 Introduction

1.1 Motivation

Deaf and hard-of-hearing (DHH) viewers receive speech through subtitles. The rest of the soundtrack is mostly lost. Captions for the deaf may name a sound in brackets, but the DCMP Captioning Key [9] tells captioners that a sound’s source “may be omitted if it can be clearly seen onscreen”, and a study of more than 715,000 YouTube videos found hand-written non-speech captions in only 5–7% of them [25]. DHH users want to know about the sounds that carry information – knocks, alarms, sirens, a baby crying [21, 32] – and want their type and timing [26].

What a viewer loses therefore depends on whether the source is visible. A siren over a visible ambulance tells the viewer nothing new. A siren over a crowd looking up *is* the scene. This project targets the second case.

1.2 Problem

Given a video, produce the same video with a side panel that shows a picture of a sound *while* the sound is heard and its source is *not* on screen, and shows nothing otherwise. Speech and music are never drawn (they have their own channels); human sounds such as laughter, coughing or a scream are. No model may be trained, and the system must work on one video at a time. For every sound this means four questions:

1. **Is there a sound, and when does it start?**
2. **What is it?** At a level a picture can show: *dog bark*, not *animal*.
3. **Can the viewer already see it?** If the thing making the sound is visibly making it, stay silent.
4. **What should the picture show?**

1.3 Research questions

From the project proposal (6 August 2026); Section 10 answers each.

- RQ1. Which types of audio information contribute most to scene understanding when presented visually?
- RQ2. What level of semantic granularity should be used to represent ambient sounds?
- RQ3. Can automatically generated visual augmentations improve accessibility beyond subtitles?
- RQ4. How should semantic audio augmentation systems be evaluated?
- RQ5. How does the proposed pipeline compare with representative audio-to-visual approaches?

1.4 Contributions

1. **A task and a metric:** a precise rule for when a picture is needed, and a per-sound score that counts found sounds, wrong pictures and their cost to a viewer (Section 3).
2. **A labelled benchmark:** 159 video clips with every sound timed and marked visible or not (Section 4).
3. **A working system** built only from open models, runnable on one video (Section 5).
4. **Results and an error analysis** that show where today’s open models stop (Sections 6–8).
5. **A complete, pre-registered development record** of every idea tried (Appendix E).

Results in brief. On a test set of 88 clips the final system finds 24 of the 65 sounds a deaf viewer would need, and half of its 48 pictures are right. Its cost to a viewer (a miss costs 4, a wrong picture 2) is 2.409 per clip. The baselines of the project proposal do worse: drawing a picture for every heard sound without looking at the video (direct audio-to-image generation, or audio captioning) costs 3.182, because only 30 % of its pictures are right, and showing nothing costs 2.955. Both differences are significant ($p < 0.001$ and $p = 0.041$). Most remaining misses are sounds the detectors barely hear under speech or music; most remaining wrong pictures show another sound, or an object that is visible but is not the one making the sound.

How the report is organised. Sections 3–5 define the task, the data and the system; Section 6 gives the results, Section 7 explains why the problem is hard for today’s models and Section 8 analyses the remaining errors; Section 9 lists every threat to validity and the limits. The appendices hold the prompts, the detection details and configuration, reproduction steps with the source of every number, a glossary of internal names, and the full development history. The code, labels and report sources are published as release v1.1.0 of the project repository (Section 10).

2 Related work

Non-speech sound for DHH viewers. Captioning practice [9] describes a sound only when the picture does not already show it; this is the visibility rule used here. Viewers want important sounds, their type and timing, and a choice [26, 1]; captions lose emotional information [30]; visual styling is welcome when it is readable and calm [10, 28]. Sound-awareness tools [20, 19, 32, 29] show that users want urgent sounds fast and are bothered by false alarms, which is why a wrong picture has a cost in the metric used here. None of this work generates pictures for ambient sound in recorded video, chosen automatically by what is already visible.

Sound event detection. Almost all sound taggers use the AudioSet label set [15] (527 classes in a tree, e.g. *Bark* under *Dog* under *Animal*). Clip-level taggers such as PANNs [23], AST [17] and BEATs [7] say *what* is heard; text-queried detectors such as CLAP [12], FLAM [35] and FlexSED [18] score a sound name frame by frame. Detection is usually scored with event-based F1 and a time tolerance (“collar”) [27, 4]; the metric used here keeps the onset tolerance but reports a cost instead of F1, in the spirit of cost curves [11]. A known weakness matters here: a quiet sound under speech or music is ranked below *Speech* or *Music*.

Audio-language and vision-language models. Audio-language models such as Qwen2-Audio [8] and Qwen3-Omni [37] answer questions about audio; the system uses two of them as “listeners”. Vision-language models such as Qwen2.5-VL [3] and LLaVA [24] answer questions about frames and are the only tool that can decide, for any sound, whether its source is visible. Their known failures matter: a preference for some answer letters [39], weak handling of negation [2], and treating a visible object as the one that is sounding (Section 8).

Sound-source localisation and audio-to-image generation. Localisation models [31, 34] find *where* in the frame a sound comes from, but barely react to off-screen or silent audio [22], so they do not answer *whether* it is visible. Audio-to-image models [33, 5, 13] draw a picture from a whole soundtrack, so they draw the scene rather than the one unseen source and cannot be told to leave a visible object out. A text-to-image model (Qwen-Image [36]) with a short, guarded prompt is therefore used, and the proposal’s direct audio-to-image baseline is represented by the same detector and generator drawing every sound, without the on-screen check (Section 6.1). Model judges [38] were tried for evaluation and failed (Appendix E.7).

3 Task and metric

3.1 Terminology

Final system the system described in this report (release v1.1.0 of the code).

Direct audio-to-image the baseline that draws every heard sound without looking at the video (Section 6.1).

Show nothing a system that never shows a picture.

Gate (on-screen check) the step that decides whether a sound’s source is visible.

Listener an audio-language model asked about a sound (“is a siren present?”, “list what you hear”).

Family a sound and its parents in the AudioSet tree (*Bark* belongs to the family *Dog*).

Span the time interval of one detected sound.

Bootstrap resampling the clips many times to see how much a difference could change by chance.

Holm correction a stricter p -value bar used when several comparisons are made at once.

3.2 Needed sounds

The unit of evaluation is one sound in one clip. For every sound the annotator records its label, its start and end (about 0.5-s precision) and three judgements:

visible the thing making the sound is *seen making it now*. A barking dog on screen is visible; a baby held quietly is not visibly crying; an alarm box on a wall is not visibly ringing.

obvious with the sound off, a viewer would still know it is happening (waves when the sea fills the frame).

importance rated as if the screen were black: 1 = steady background noise; 2 = an event one can say in a sentence; 3 = danger or a key moment (siren, alarm, gunshot, glass, baby crying, scream).

A sound is **needed** if it is neither visible nor obvious and has importance 2 or 3. *Example*: a police siren is heard while the camera shows a crowd looking up – needed. When the police car drives into view with its lights on, the siren becomes visible – not needed. Every heard sound is labelled, because a picture over an unlabelled moment counts as wrong.

3.3 Scoring a picture

Pictures are scored as they are shown on screen. A picture **hits** a needed sound if it is of the same family and *starts* between 0.5s before and 1s after the sound starts; each sound takes at most one picture. Every other picture is **wrong**, of one of three kinds:

on screen it shows a sound whose source is visible (the viewer could see it);

other sound it shows the wrong sound – something else is heard there (a gunshot picture over footsteps), or it is a late second picture of a long sound already shown;

nothing there is no labelled sound there at all.

“Wrong” is therefore strict: it includes redundant pictures, not only false ones.

3.4 Viewer cost

The main score is the cost per clip,

$$C = \frac{4 \cdot \text{\#missed needed sounds} + 2 \cdot \text{\#wrong pictures}}{\text{\#clips}}, \quad (1)$$

lower is better. A miss costs twice a wrong picture, so one right picture pays for two wrong ones, and a picture is worth showing when it is right at least one time in three. These weights are an assumption, not a measurement: they were chosen in September for another purpose and became the main score only after F1 showed no difference between systems. Section 6.4 therefore also reports other weights, including a version in which an on-screen picture costs only 1 or 0 (it is less harmful than a picture of nothing):

$$C_w = \frac{4 \cdot \text{\#missed} + w \cdot \text{\#on screen} + 2 \cdot \text{\#other sound} + 2 \cdot \text{\#nothing}}{\text{\#clips}}, \quad w \in \{0, 1, 2\}. \quad (2)$$

Why not F1? F1 counts a miss and a wrong picture the same, while the gate’s value lies almost entirely in the wrong pictures it removes; F1 is still computed by the scorer.

3.5 How decisions were made

Comparisons are paired by clip and tested with a paired clip bootstrap. A change to the system was accepted only if it passed a rule written down *before* its run, on the development set: at least as many hits, at most two extra wrong pictures per extra hit, and a lower cost (or, losing at most three hits, at least three fewer wrong pictures per hit lost). Rules about the audio alone could first be screened on a separate set of AudioSet clips. The test set was read only after a change was accepted, and only reported. With 88 test clips the smallest cost difference the design can detect is about 0.19 per clip, so most single changes could only be judged on the development rules.

4 Data

All results use two sets of short video clips (10–20s), labelled per sound as in Section 3.2:

Development set 71 clips, 58 needed sounds. Every design decision was made on this set.

Test set 88 clips, 65 needed sounds. Read only after a change was accepted.

The clips come from Wikimedia Commons, the Internet Archive, UnAV-100 [16], VGGSound [6], YouTube (city walks, markets, home, transport, incidents, film scenes) and the AudioSet-Strong evaluation split. New clips were assigned to development or test by a hash of the clip name or by balanced pairs, never by looking at results; Appendix E.1 describes how the sets grew. Most clips are for research use only.

Table 1: The two evaluation sets. Scenario counts: *unseen* (an ambient source is off screen), *mixed* (some sources visible, some not), *seen* (all sources on screen), *none* (speech or music only).

Set	Clips	Labelled sounds	Needed sounds	Clips with a needed sound	Unseen / mixed / seen / none
Development	71	154	58	38	24 / 16 / 23 / 8
Test	88	154	65	43	21 / 23 / 22 / 22

About half of the clips have no needed sound; on those, the best possible behaviour is to show nothing. Three needed development sounds (a clang and two golf-club strikes) belong to labels the system never draws, so they are misses for every system.

Label quality. All labels are the author’s; a second annotator was planned but did not return labels, so inter-annotator agreement is unknown (on the earlier clip-level labels, a blind relabelling by the author agreed 78% of the time, Cohen’s $\kappa = 0.60$). The labelling tool showed detector suggestions on many clips, and a blind check of pictures counted as wrong found that about 1 in 6 is a real off-screen sound the labels miss. These and other caveats are listed in Section 9.

Screening set. 415 clips of the AudioSet-Strong evaluation split, which already have timed labels, were used to screen rules about audio alone (for example: how often is a detection that no listener names a real sound?). AudioSet labels only a few sounds per clip, so this set was never used to accept a change.

5 The system

This section describes the final system as the code runs it (`config.use_shipped()` in `config.py`). Appendix B gives every step and threshold.

5.1 Overview

Figure 1 shows the data flow and Figure 2 an output. For one video:

1. **Hear candidates.** Two sound detectors listen to the whole clip: BEATs (a tagger for the 527 AudioSet classes) and FlexSED (a detector that is asked one sound name at a time). Anything either hears strongly enough for at least 0.3s becomes a candidate with a start and an end.
2. **Check candidates.** A candidate that only one model claims is removed. Weak sounds that the detectors almost missed get second opinions from two listeners (Qwen3-Omni and Audio Flamingo Next) and a third audio model, the Detect Any Sound Model (DASM).
3. **Witness rule.** A remaining weak candidate survives if one of three witnesses backs it: DASM clearly hears it at that time; or both listeners list it; or one listener lists it and the vision model says the sound fits the scene.
4. **On-screen check (the gate).** A vision-language model (Qwen3.8-27B) looks at frames around the sound. If the thing making the sound is visibly making it, no picture is shown.
5. **Clean the display.** Repeats close together become one picture; a listener decides whether two same-family pictures up to 8s apart are one continuing sound; a picture is dropped when its event is visibly happening.
6. **Draw.** A short, guarded prompt is written for each remaining sound; Qwen-Image draws it; the vision model checks that the picture shows the intended thing and it is redrawn up to five times.
7. **Compose.** Pictures appear in up to three fixed slots beside the video while their sound plays (at least 1.5s).

5.2 Models at a glance

5.3 Detection

The two detectors complement each other: BEATs is strong on clear sounds, FlexSED hears quiet sounds under speech and music far better. Most of the development effort went into deciding which weak candidates to keep. The listeners are asked about each weak candidate in two ways – a yes/no question (“Is the sound of a siren present in this recording?”) and an open list (“List every distinct non-speech sound you hear”). On the screening set, a detection that neither listener names is a real sound only 22 % of the time, one named by one listener 37 %, and one named by both 62 %; the witness rule is built on this. Speech and music are never drawn; human sounds and all other sound families with a clear source are (Appendix B).

5.4 The on-screen check

For every detected sound, the gate cuts its time span into stretches of about 5s, shows the vision model six frames from each stretch and asks three questions (exact prompts in Appendix A):

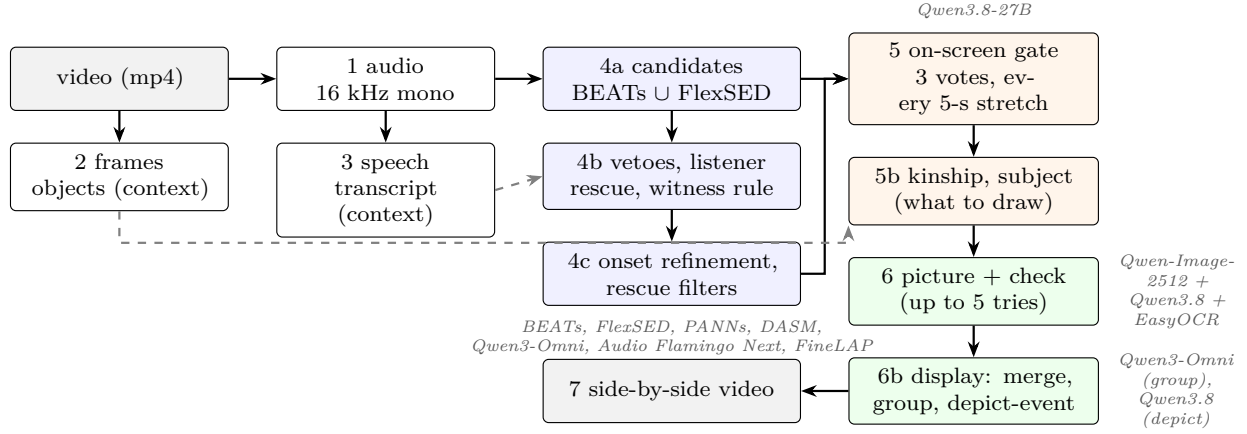


Figure 1: The final pipeline. Blue: detection (stage 4); orange: the on-screen decision (stage 5); green: pictures and display. Dashed: context only (objects and transcript never decide on their own). The scene check of the witness rule in stage 4b also uses Qwen3.8-27B.

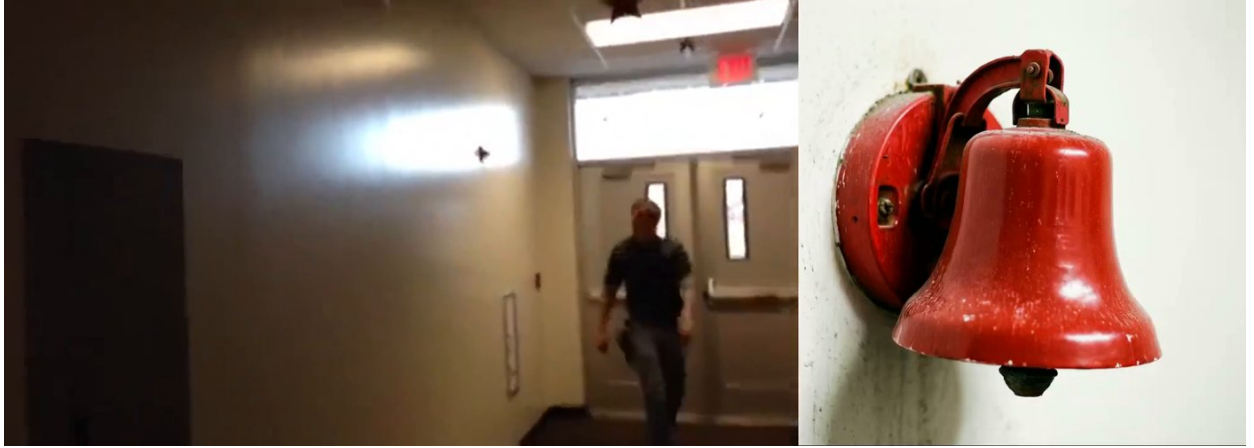


Figure 2: Output of the final system on a development clip (AudioSet-Strong, a corridor). A fire alarm rings from 9.0s; the alarm bell itself is not in the frame. The system detects *Alarm*, the on-screen check finds no visible source, and the panel on the right shows the generated picture from 8.9s while the alarm sounds. In the scoring this is a hit.

1. **Name it:** “Name the thing in these frames that is making that sound, or answer: nothing.” If the answer shares a word with the label (“church bell” for *Bell*) this vote says visible; otherwise a text-only follow-up asks whether that thing could make the sound.
2. **Choose:** “(a) you can see it happening on screen, or (b) it is not visibly happening”, asked in both orders; if the two answers disagree the vote abstains.
3. **Describe:** a one-sentence description of the frames, checked against the sound in the same way.

A stretch counts as visible if most votes say so, and the sound is silenced only if *every* stretch is visible. This leans toward showing: a source that leaves the frame for one stretch keeps its picture.

5.5 Pictures and display

The picture prompt is built from the detected label, not from free text: the vision model may add at most two qualifier words to the source’s name, word lists remove people and places, and an action is drawn with its maker (a *bark* becomes a barking dog). A fixed tail asks for one large subject on a plain white background. Qwen-Image-2512 draws it at 1024×1024; the vision model then gets a multiple-choice question about what the picture shows, and a picture that fails (or contains

Table 2: Stages, models and key parameters of the final system (in the order of `src/pipeline.py`).

#	stage	model / tool	key parameters
1	audio	ffmpeg	mono WAV, 16 kHz (<code>SAMPLE_RATE</code>)
2	frames, objects	OWLv2 (<code>owlv2-base-patch16-ensemble</code>)	6 frames, threshold 0.20; context only, never silences a sound
3	speech	faster-whisper <code>base</code> , int8	transcript used only to ask whether people react to a sound; never drawn
4	sound events	BEATs iter3+ AS2M; FlexSED; PANNs CNN14; DASM; Qwen3-Omni-30B-A3B; Audio Flamingo Next; FineLAP	16 ordered steps, Table 5
5	on-screen gate	Qwen/Qwen3.8-27B, greedy, thinking off	6 frames per ≤ 5 -s stretch, 3 votes, majority; silenced only if every stretch is visible
5b	subject	Qwen3.8-27B + word lists	2–6 word phrase; speech text never used
6	picture	Qwen/Qwen-Image-2512	1024^2 , 50 steps, true CFG 4.0; check-and-redraw ≤ 5 tries
6b	display	Qwen3-Omni (group), Qwen3.8 (depict)	merge gap 2.5 s, min. 1.5 s on screen, ≤ 1 s past the end, group ≤ 8 s, 3 slots
7	composite	ffmpeg libx264	720-px panel, 25 fps

text) is redrawn, up to five times. In a blind glance test by the author, Qwen-Image pictures were recognised 26 of 54 times against 14 for the earlier FLUX.1 pictures (+0.22, interval [+0.07, +0.37]). On screen, repeats of one sound less than 2.5 s apart become one appearance, every picture stays at least 1.5 s and at most 1 s after its sound ends, and at most three pictures are shown at once.

The scored numbers come from a harness that replays the pipeline’s own functions on cached model outputs; an independent re-export of every decision reproduces them exactly. The live pipeline differs in four small ways that do not change when a picture starts (Appendix B).

6 Results

6.1 Baselines

The project proposal (Section 7, “Baseline systems”) names the systems the method is compared against. All are run on exactly the same clips and scored with exactly the same rules:

Direct audio-to-image generation generates a picture for every sound it hears, without looking at the video. It uses the same detector and the same picture model as the final system, so the only difference is the on-screen check and the display rules that depend on it. This isolates the contribution the proposal asks about: reasoning about what the video already shows. (Published audio-to-image models draw a whole scene from the soundtrack and cannot be scored per sound; Section 2.)

Audio captioning describes the same sounds in text instead of pictures. Under the per-sound score a caption is right or wrong at exactly the same moments as the audio-to-image picture, so it has the same score; the difference between text and pictures can only be measured with viewers (Section 9).

Show nothing no augmentation at all: the situation of a viewer with subtitles only.

The supervisor also suggested a generator conditioned on the video and audio that would “generate what is not seen” (for example MiniMax-H3). It produces whole new video, which the per-sound score cannot evaluate, so it was not run as a baseline (Section 9).

6.2 Main result

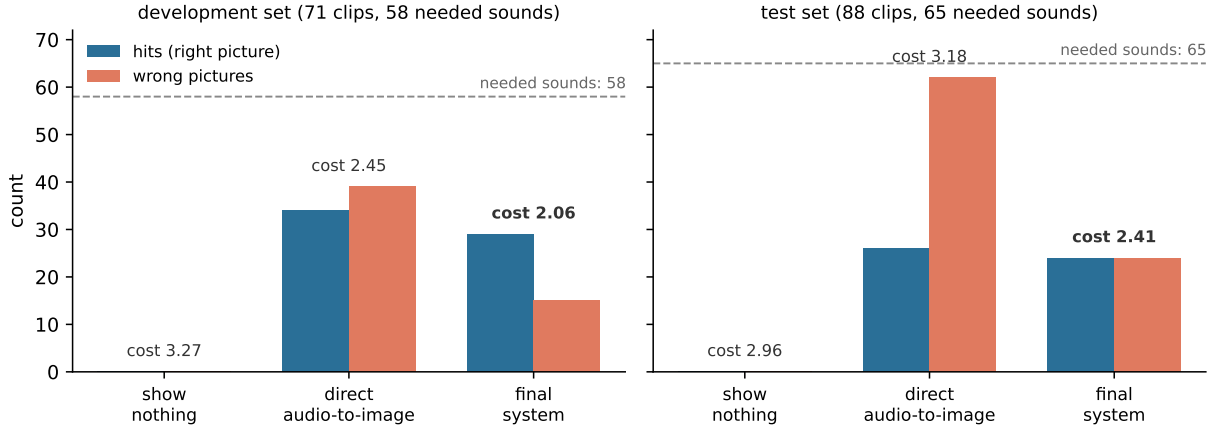


Figure 3: Hits (right pictures of needed sounds) and wrong pictures for each system, with the cost per clip above each group (lower is better). Drawing every heard sound finds a few more needed sounds but shows two to three times as many wrong pictures; the final system keeps most of the hits and removes most of the wrong pictures.

Table 3: Main result. Wrong pictures are split into on screen / other sound / nothing. Differences are in cost per clip, with a 95 % interval from a paired clip bootstrap (10,000 draws) and a two-sided p -value.

Set	System	Hits	Wrong	Cost	Final system minus this	
Development (71 clips)	show nothing	0 / 58	0	3.268	-1.211 [-1.831, -0.620]	$p < 0.001$
	direct audio-to-image ^a	34 / 58	39 (20/14/5)	2.451	-0.394 [-0.789, +0.056]	$p = 0.088$
	final system	29 / 58	15 (6/7/2)	2.056		
Test (88 clips)	show nothing	0 / 65	0	2.955	-0.545 [-1.068, -0.023]	$p = 0.041$
	direct audio-to-image ^a	26 / 65	62 (16/32/14)	3.182	-0.773 [-1.045, -0.500]	$p < 0.001$
	final system	24 / 65	24 (4/15/5)	2.409		

^a Audio captioning has the same score (same decisions, shown as text).

On the test set the final system finds 24 of 65 needed sounds and half of its pictures are right. Drawing every heard sound finds 26, but only 30 % of its pictures are right, and on the test set it is no better than showing nothing (+0.227 per clip, [-0.386, +0.818], $p = 0.49$). The final system is significantly cheaper than both: by 0.773 per clip against direct audio-to-image generation ($p < 0.001$) and by 0.545 against showing nothing ($p = 0.041$). On the development set the direction is the same; the difference from direct audio-to-image is smaller there (-0.394, $p = 0.088$) because the detector heard more of the needed sounds on that set.

6.3 What the statistical tests support

Tested and significant. On the test set the final system costs less than direct audio-to-image generation and audio captioning ($p < 0.001$) and less than showing nothing ($p = 0.041$). On the development set it costs less than showing nothing ($p < 0.001$).

Not significant. On the development set, the difference from direct audio-to-image generation ($p = 0.088$).

Caveat. The test set was read after each accepted change during development, so the test numbers are a report on a seen set rather than a blind confirmation (Section 9).

6.4 Sensitivity to the cost of a wrong picture

The weight of a wrong picture relative to a missed sound (β) is the one number in the metric that only deaf and hard-of-hearing viewers can set. A viewer study to measure it was designed but not run, so the main analysis assumes $\beta = 2$. Figure 4 shows how the conclusions change if β is different.

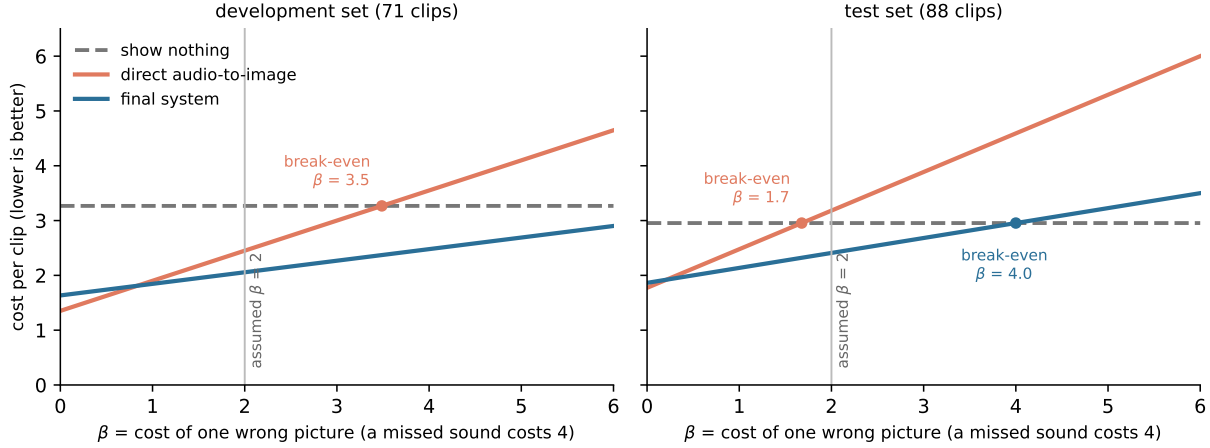


Figure 4: Cost per clip as the cost of a wrong picture (β) varies, with a missed sound fixed at 4. Dots mark where a system breaks even with showing nothing. On the test set the final system is cheaper than showing nothing for every $\beta < 4.0$, i.e. until a wrong picture hurts as much as a missed sound (on the development set until $\beta = 7.7$, off the chart). Direct audio-to-image generation breaks even already at $\beta = 1.7$ (test) and 3.5 (development). The final system is cheaper than direct audio-to-image generation for every β above 0.2 (test) or 0.8 (development).

The conclusion therefore does not depend on the exact value of β . Drawing every sound is only preferable if wrong pictures are almost free (β below 0.2 on the test set); the final system beats showing nothing as long as a wrong picture is judged less harmful than a missed sound. A second check weighs only on-screen pictures less (Eq. 2): with $w = 1$ or $w = 0$ the final system costs 1.972 or 1.887 (development) and 2.364 or 2.318 (test).

6.5 Where the errors come from

Figure 5 breaks down the misses and wrong pictures of the final system, from its logged decision trail. Most misses are sounds that were heard as candidates and then removed by a later rule – most often because the listeners did not confirm a weak detection. Each of these rules was accepted because, overall, it removed more wrong pictures than real ones; the sounds it removes are the price. Six needed sounds in each set were heard by no detector. Most wrong pictures on the test set show another sound (15 of 24). Section 7 explains why these errors are hard to remove.

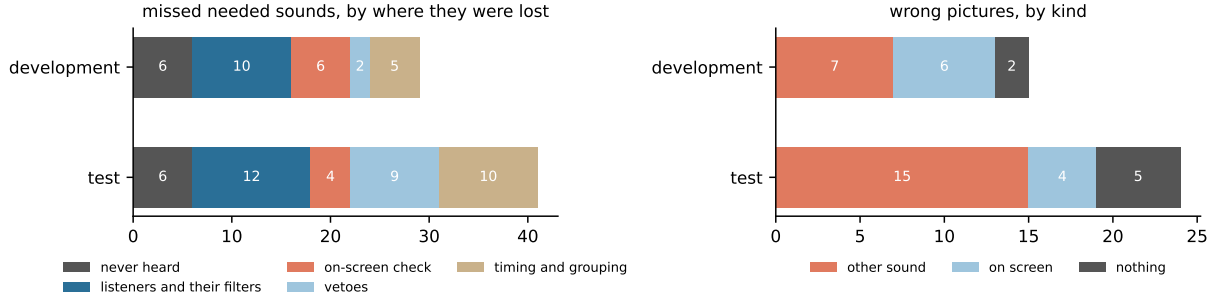


Figure 5: Misses of the final system by the step where the sound was lost (left), and wrong pictures by kind (right).

7 Why the problem is hard for today’s models

A reader may expect that modern models solve this task directly: a sound tagger hears the siren, a vision model says whether an ambulance is on screen, an image model draws it. This section explains, step by step, why that does not happen. “Hard” here means hard for open models of up to about 70 GB, chained without training; a much larger vision model (Qwen3-VL-235B) was not tested, and fine-tuning was outside the design. Each subsection states what one might expect, what actually happens, and the evidence.

7.1 Hearing the sound at all

Expected: a modern tagger hears a hammer or a door. *Actually:* under speech or music it often does not. The tagger’s own score for *Speech* or *Music* is higher than for the quiet sound, so the sound never becomes a candidate. Even with every later stage made perfect, 12 of 55 needed development sounds stay missed: 5 are heard by no model at all and 7 are too faint or too badly timed (Figure 6). Removing the speech first with a source separation model made it worse: on DCASE events, recall under speech fell from 9.5 % to 4.8 %, because the detectors were never trained on separated audio.

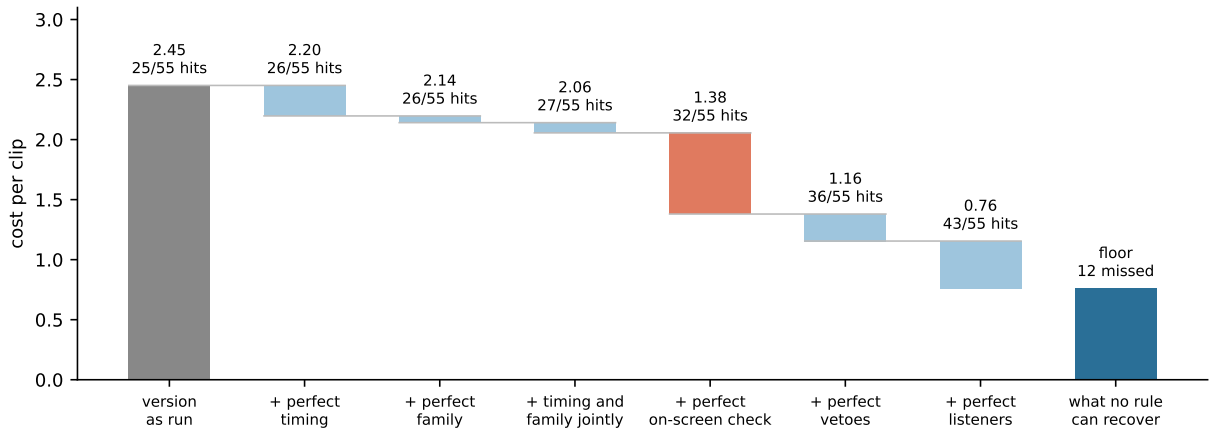


Figure 6: Ceiling analysis on a slightly older version of the system (development set, 55 needed sounds). Each bar makes one more stage perfect, using the labels, on the same detections. The largest single step is the on-screen check. Even with every stage perfect, 12 needed sounds remain missed (right): the detectors never hear them clearly enough.

7.2 Knowing whether a weak detection is real

Expected: an audio-language model can confirm “is there a siren?”. *Actually:* its answers are only weak evidence. On the screening set, a weak detection that neither listener names is a real sound 22 % of the time, one named by one listener 37 %, and one named by both 62 % (Figure 7). Every way to add more sounds found real ones but paid two to four wrong pictures per new hit: the best such rule gained 3 hits on the development set (31 hits, 26 wrong) and lost on the test set (25 hits, 35 wrong). Nothing later in the chain can tell a right off-screen name from a wrong one.

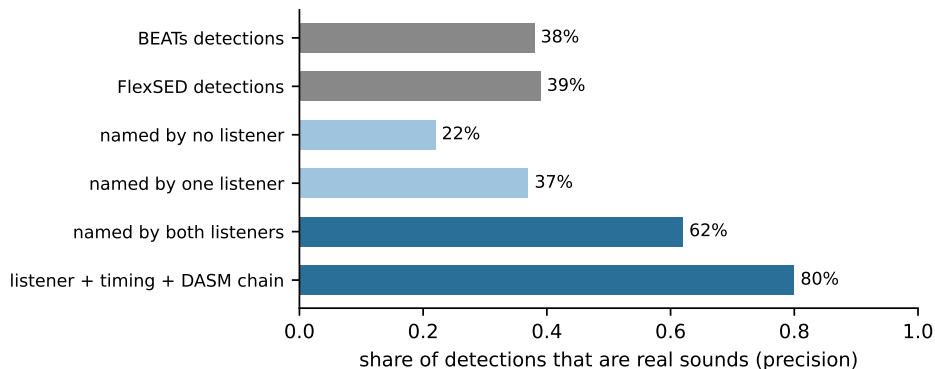


Figure 7: Share of detections that are real sounds, on the screening set (415 AudioSet-Strong clips with timed labels). Raw detections are right less than 40 % of the time; agreement between models helps, but a sound only one listener names is right just over a third of the time.

7.3 “It is there” is not “it is making the sound now”

Expected: a vision-language model can answer “is this bell ringing?”. *Actually:* it answers “is there a bell?”. A church tower on screen silences a bell heard off screen; perched birds are taken as the source of a call made elsewhere. This explains 23 of 36 gate errors in the audit. The opposite failure is also large: asked to name the source, the model answers “nothing” on 42 % of the stretches the labels call visible (a hand, the footsteps of the person holding the camera). The problem is old: a CLIP-based check in August reached 49.3 % accuracy and still augmented 71 of 103 clips whose sources were visible. Going from a 7-billion to a 27-billion-parameter vision model moved balanced accuracy only from 0.61 to 0.62.

7.4 Every fix to the on-screen check trades one for one

Expected: adding an object detector, a segmentation model, a crop of the source, a model of audio-visual synchrony or a sound-localisation model should help. *Actually:* each one silenced more on-screen sounds only by silencing about as many needed ones (Figure 8). Removing two wrong pictures saves $2 \times 2 = 4$ cost units; losing one needed sound costs 4: no gain. Localisation models also react to off-screen and silent audio, so they answer *where* a sound comes from, not *whether* its source is visible.

7.5 Answers follow the shape of the question

Expected: a clear question gets a clear answer. *Actually:* given options (a) and (b), the 27B model picks (a) 88 % of the time in one test; asked yes/no, it leans to “no”; asked for free text, its answer can be cut off. The final system therefore reads how strongly the model prefers “yes” over “no” at the first word, minus the same for the opposite question. This removes the habits but reveals the real weakness: as a signal for separating visible from needed sounds on the labelled development stretches, it reaches an AUROC of only 0.647 (0.5 is a coin flip, 1 is perfect). The habits were hiding a weakness, not causing it.

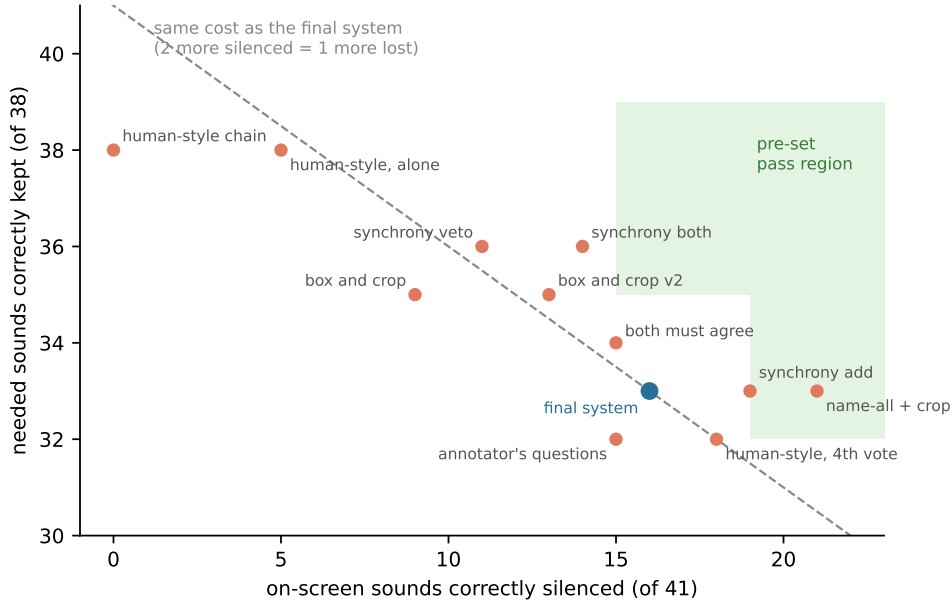


Figure 8: Every variant of the on-screen check that was tried, scored on the development sounds labelled by the author (41 on-screen sounds, 38 needed sounds). The dashed line has the same cost as the final system: silencing two more on-screen sounds is worth losing one needed sound. All variants lie near this line, and none reached the pass region fixed before the tests.

7.6 Drawing one unseen source, not a scene

Expected: an audio-to-image model can draw what is heard. *Actually:* such models draw the whole soundtrack as a scene and cannot be told to leave out the dog that is already on screen. The system therefore draws from the label with a guarded text prompt – and even then the author recognised a picture at a glance only 26 of 54 times (14 of 54 with the earlier model). A richer prompt reached 32 but invented objects (a thud drawn as a door).

7.7 Measuring success is hard too

Expected: a model judge can score the output. *Actually:* a reference written by the system’s own models rewards the system; one written by independent models agreed with the human “nothing is missing” label on 55 of 100 clips, which is chance; and the best judge scored a picture and a text tag the same (Appendix E.7). Existing labels do not help either: AudioSet names only a few sounds per clip, so on AudioSet labels the system looked worse than showing nothing (1.650 against 0.986 per clip), although on full labels it is clearly better. Even full human labels miss some sounds: about 1 in 6 “wrong” pictures is a real off-screen sound, and 14 borderline development sounds sit one judgement away from the other class.

7.8 The data is rare

Clips in which an ambient sound is heard but its source is not seen are uncommon in public video. Four different sourcing routes all yielded only 15–20 % usable clips, which is why the benchmark has 159 evaluation clips rather than the planned 300, and why every comparison has wide intervals.

8 Error analysis

This section looks at the remaining errors of the final system in detail, using its logged decision trail, per-item error lists and an audit of the on-screen check (sources in Appendix C).

8.1 Ceiling analysis

Table 4 gives the numbers behind Figure 6: each row makes one stage perfect on its own. The analysis needs saved pictures, so it was run on a slightly older version of the system (25 hits and 27 wrong on the development set, 55 needed sounds); the proportions, not the exact numbers, are the point.

Table 4: One stage made perfect at a time (development set, slightly older version).

Stage made perfect	Hits	Wrong	Cost (from 2.451)
On-screen check	30	18	1.915
Listeners and their filters	34	24	1.859
Vetoes and filters	30	27	2.169
Timing	26	20	2.197
All stages perfect	43	3	0.761

The on-screen check and the listeners are the two largest losses. Even with every stage perfect, 12 of 55 needed sounds stay missed: five are heard by no model at all (a hammer, an explosion, a clang and two golf-club strikes) and seven are too faint or too badly timed.

8.2 The on-screen check in detail

The audit of the gate’s decisions found two kinds of error. The more common one (23 of 36 gate errors) is *presence taken for production*: an object in the frame is taken as the object making the sound. This follows from how the vote is built. Two of the three questions effectively test presence – naming an object, and asking whether that object *could* make the sound – and only the a/b question asks whether the sound is visibly happening, so the majority tests presence too. The second error is the opposite: the name question answers “nothing” on 42 % of the stretches the labels call visible, when the source is small, hidden, or its action is fine.

8.3 Label strictness

In two blind checks, about 1 in 6 pictures counted as wrong was a real, wanted, off-screen sound that the labels do not list, and several others were real sounds with the source on screen. Absolute costs are therefore somewhat pessimistic; the comparison between systems is not affected, because the same labels score all of them.

8.4 Worked example

Clip `bell_miami` (development set) has one needed sound: a bell from 0.2 to 14.5s, labelled not visible. The detectors find it. The gate cuts the 14-s sound into three stretches. In each, the name question answers “church bell” (it shares the word *bell*, so this vote says visible), the a/b question says not visibly happening, and the description vote says visible. Two of three votes in every stretch: the sound is silenced and the needed bell is missed. The frames show the church; they do not show the bell ringing. The one question that asked about the action was right, and it was outvoted by two questions that measure presence.

9 Limits, threats to validity and future work

This section separates three things: what the approach itself cannot do, what the evaluation cannot show, and what could make the reported numbers wrong. Section 7 explains the first in detail.

9.1 Limits of the approach

- **Detector floor.** About one needed sound in five cannot be found with today’s open detectors, mostly because it is quiet under speech or music. No later rule can recover it.
- **Presence is not production.** The on-screen check confuses an object being visible with the object making the sound; every alternative tried traded one error for the other.
- **No stage can tell a right off-screen name from a wrong one,** so adding more candidate sounds adds wrong pictures at about the same rate as hits.
- **Offline only.** A 15-s clip needs about five minutes of GPU time (one NVIDIA H200) before the first picture is drawn, and each picture try takes about 20 s. The system cannot caption live television.
- **Design choices.** No model is trained; speech and music are never drawn; one picture is shown per sound, without words.

9.2 Limits of the evaluation

- **No viewer study.** The cost model assumes that a wrong picture harms a viewer half as much as a missed sound. No deaf or hard-of-hearing viewer has confirmed this, or that a generated picture is understood faster than a caption in brackets. The metric is therefore a model of the viewer, not a measurement; Figure 4 shows that the main conclusion holds for any weight up to 4.
- **One annotator.** All labels are the author’s; inter-annotator agreement is unknown.
- **Strict “wrong”.** It counts redundant and late pictures, and about 1 in 6 “wrong” pictures is a real off-screen sound missing from the labels.
- **Pictures checked by the author only.** The picture result rests on one blind glance test by the author; the automatic picture check was validated on known cases, not against a human rater.

9.3 Threats to validity

- T1. The test set was seen.** It was read after each accepted change, about 19 times in the last week (its 60 older clips 11 times before that). Twice a choice was influenced by a test read; both were undone. About a third of the development gain carried over to the test set (Appendix E), which is consistent with selection on the development set. The test numbers are a report on a seen set, not a blind confirmation.
- T2. Small sets.** With 71 and 88 clips, one or two pictures can decide a development comparison, and the test set detects only cost differences of about 0.19 per clip or more.
- T3. Label changes.** The author re-checked the visibility of 14 disputed development sounds blind (3 changed; 0 of 20 random controls changed), which may favour the system; test labels were never changed. Newer clips were labelled with detector suggestions visible, and 19 AudioSet clips with the AudioSet labels visible.
- T4. One baseline not run.** The video-conditioned generator suggested by the supervisor (for example MiniMax-H3) creates new video; the per-sound score cannot evaluate it, so it was not compared.
- T5. Scored versus live system.** The scoring harness and the live pipeline differ in four small ways that do not change when a picture starts; one of them (the direction of a family rule) was not re-scored (Appendix B).

9.4 Future work

Each item addresses one of the limits above.

1. **Measure the viewer’s costs:** a study with 5–10 DHH viewers (“helped / hurt / neither” per clip) that estimates the weight of a wrong picture instead of assuming it.
2. **A second annotator** for 30 clips (the packet exists), to measure label agreement.
3. **The presence-versus-production question:** a much larger vision model, a comparison of the frame at the sound’s onset with a frame just before it, or an external box around the named object.
4. **A detector that hears sounds under speech and music**, tested on the unused set of 422 AudioSet clips.

10 Conclusion

A deaf viewer misses the sounds whose source is not on screen, and no existing channel supplies them automatically. This project built a system that does, from open models with no training, together with a labelled benchmark and a metric that prices what a viewer loses. On the test set the final system finds 37% of the needed sounds, half of its pictures are right, and its cost is significantly lower than both baselines of the proposal: direct audio-to-image generation ($p < 0.001$) and showing nothing ($p = 0.041$). The remaining errors sit where today’s open models stop: sounds the detectors barely hear under speech and music, and a vision model that takes “the object is there” for “the object is making the sound”.

Answers to the research questions.

RQ1 – which audio information matters most?

Discrete, sourced events whose source is off screen: sirens, alarms, bells, a baby crying, glass, a knock. The benchmark’s importance rule (an event one can say in a sentence, or danger) captures them; steady place noise (wind, traffic hum) is excluded by design, and it is also where detectors, gate and labels disagree most. Whether DHH viewers rank sounds the same way is untested.

RQ2 – what granularity?

Decide and score at the level of the AudioSet *family* ($\text{depth} \geq 1$), draw the specific source. Family matching was the one de-duplication and scoring rule that generalised without a threshold; the picture names the maker (a barking dog, not “bark”) from a hand table, which beat generic VLM wording 7/7 to 3/7.

RQ3 – can generated augmentations improve accessibility beyond subtitles?

Not shown. What is shown is that the system adds information no subtitle carries (off-screen sounds) at a modelled cost below showing nothing, and that its pictures are recognised at a glance about half the time. A study with viewers is the necessary next step.

RQ4 – how to evaluate?

Per sound, against human labels, with the error split by cause and a cost that prices a miss against a wrong picture, reported over a range of weights; decisions pre-registered on a development set. Automatic protocols with model-written references or model judges failed: they are circular, decide visibility at chance, or cannot tell a picture from a text tag.

RQ5 – comparison with audio-to-visual approaches?

Against the proposal’s direct audio-to-image baseline (same detector and picture model, every heard sound drawn), the final system costs 0.773 less per clip on the test set ($p < 0.001$): it has two fewer hits but 38 fewer wrong pictures. Audio captioning scores the same as direct audio-to-image under the per-sound metric; whether pictures beat text needs viewers.

Code and data availability. The code, the per-sound labels, the scoring harness, the logged decisions of the final system and the sources of this report are published as release v1.1.0 of <https://github.com/adamgavriely/MscFinalProject> [14]. The video clips are not redistributed; Appendix C explains how to reproduce every number.

A Prompts used in the system

All questions are asked with greedy decoding and, for Qwen3.8-27B, thinking off. $\langle \text{label} \rangle$ is the family name, lower-cased and cut before a comma or bracket. Source: `src/stage5_cross_modal_analysis/reason.py`, `benchmark/listener_round.py`, `benchmark/gold/listener_variants.py`, `src/stage6_visual_augmentation/group.py`, `depict.py`, `verify.py`, `gen_screen.py`.

Gate: name vote

“These frames are from the moment a sound of $\langle \text{label} \rangle$ was heard. Name the thing in these frames that is making that sound. Answer with a short noun phrase of at most 5 words. If nothing that could make that sound is visible in these frames, answer exactly: nothing.”

Gate: follow-up (text only)

“Sound heard: $\langle \text{label} \rangle$ / Thing visible in the video: $\langle \text{named} \rangle$ / Could that thing be what is making that sound? Answer yes or no.”

Gate: a/b vote (asked in both orders)

“These frames are from the moment a sound of $\langle \text{label} \rangle$ was heard. Judge from the frames alone. (a) you can SEE $\langle \text{label} \rangle$ happening on screen – the source is in the frames and visibly making that sound (b) $\langle \text{label} \rangle$ is not visibly happening in these frames. Answer with the letter only.”

Gate: describe vote

“Describe what is happening in these frames in one sentence. Name what you see and what it is doing. No more than 25 words.”

Speech context (text only, both orders)

“... Are they (a) reacting to that sound or talking about it, or (b) not referring to that sound?”

Scene check (witness rule; logit read)

“Could the sound of $\langle \text{label} \rangle$ plausibly be heard in this scene? Answer yes or no.” and the twin
“Could the sound of $\langle \text{label} \rangle$ NOT plausibly be heard in this scene? Answer yes or no.”

Listener yes/no (Qwen3-Omni, AF Next)

“Is the sound of $\langle \text{family} \rangle$ present in this recording? Answer yes or no.”

Listener inventory

“List every distinct non-speech sound you hear in this recording, one per line, most prominent first.” (64 new tokens)

Grouping (Qwen3-Omni, both orders)

“You hear a $\langle \text{lab} \rangle$ sound near the start and again near the end of this recording. Is the $\langle \text{lab} \rangle$ near the end $\langle o_1 \rangle$, or $\langle o_2 \rangle$? Answer with exactly one word: $\langle w_1 \rangle$ or $\langle w_2 \rangle$.” with the options “the same continuing sound as at the start (for example one alarm or engine with a short pause)” and “a new, separate event (for example a second bark, knock or shot)”.

Depict-event

“Is “ $\langle \text{event} \rangle$ ” visibly happening in these frames? Answer yes or no.” and “Do these frames show no “ $\langle \text{event} \rangle$ ” happening? ...”; then (text only) “Could the sound of $\langle B \rangle$ be mistaken for the sound of $\langle A \rangle$?” in both directions.

Picture check

“Look at this picture. What is the main thing shown in it? Choose exactly one option. ... Answer with the letter only.”

Picture prompt tail

“, the whole thing fully in frame, caught at the moment it makes the sound with its visible effect, one large subject filling the picture, plain white background”; negative prompt: scenery, landscape, background scene, room interior, street, buildings, sky background, text, letters, watermark.

B System details and configuration

This appendix gives the parts of the final system that the main text summarises: every step of the detection stage, the label filter, how the scored numbers relate to the live pipeline, and every configuration value.

B.1 Detection (stage 4) in full

Stage 4 is where most of the development effort went. It turns the audio into a list of timed candidate sounds, each with a family label and a confidence. The entry point is `detect_events` (`src/stage4_audio_event_detection/__init__.py`); unions and vetoes are in `fuse_flexsed`, the filters on rescued spans in `filter_rescued`.

Models

BEATs

iter3+ fine-tuned on AudioSet-2M (checkpoint `BEATs_iter3_plus_AS2M_finetuned_on_AS2M_cpt2`). It scores a whole window, so it is run on 2-s windows every 0.25 s; the first two seconds are mirror-padded so a sound at the very start can be heard; each window’s score is stamped 0.5 s before the window ends, so a detected start is never earlier than the true one.

FlexSED

an open-vocabulary, frame-level detector [18]: a self-supervised audio encoder plus a CLAP text encoder (`laion/clap-htsat-unfused`), trained on AudioSet-Strong. It is asked one plain query per family over the 215 *depictable* families (`benchmark/gold/depictable_vocab.json`), 25 frames per second. It hears quiet sounds under speech and music far better than BEATs.

PANNs CNN14

(sound-event version, 32 kHz) used only as a clip-level veto.

DASM

Detect Any Sound Model (ACM MM 2025; weights `CPF2/detect_any_sound`), queried with “sound of ⟨family⟩” over the same 215 families, 50 frames per second. Used as a witness and a vote.

Listeners

`Qwen/Qwen3-Omni-30B-A3B-Instruct` and `nvidia/audio-flamingo-next-hf`. Each is asked a yes/no question (“Is the sound of ⟨family⟩ present in this recording?”, scored as the logit of *yes* minus the logit of *no*) and an open inventory (“List every distinct non-speech sound you hear in this recording, one per line, most prominent first.”). An inventory line names a family if it matches its name, an AudioSet synonym, a parent or a child, or else has sentence-embedding cosine > 0.6 (`all-mpnet-base-v2`). The audio cut is the candidate span ± 1 s, at least 1 s long.

FineLAP

(`AndreasXi/FineLAP`, MIT) a frame-level audio–text model used as a last veto on rescued sounds.

Steps in execution order Table 5 gives the 16 steps. Two words are used throughout: a *rescued* span is one added by a listener after the detectors missed it; a *twinned* span is one both detectors raised.

Table 5: Stage 4 of the final system, step by step.

#	step	rule	variable(s)
1	BEATs spans	a span of any class where the window score ≥ 0.175 ; kept if ≥ 0.3 s long; confidence = peak	AED_THRESHOLD 0.175, AED_MIN_DUR 0.3
2	FlexSED spans	same extractor on FlexSED frames at bar 0.8, ≥ 0.3 s	FLEXSED_BAR 0.8
3	twin union	a FlexSED span with a same-family BEATs span within 1 s is merged into it (earlier start kept; confidence = the stronger side, rescaled); otherwise kept as FlexSED-only	TWIN_MAX, UNION_START min
4	mirror veto + listener keep	a BEATs-only span is dropped if FlexSED’s top query in its window is another family ≥ 0.7 while its own family is < 0.4 ; put back if the Qwen listener accepts it and an inventory names it	MIRROR_VETO 0.7, KEEP_NEEDS_V4
5	masked-weak veto	a BEATs-only span with confidence < 0.5 under speech or music (≥ 0.3) is dropped unless a listener accepts it	MASKED_WEAK_VETO
6	DASM clip veto	a span whose family DASM never reaches 0.0839 anywhere in the clip is dropped unless a listener keeps it	DASM_CLIP_VETO
7	witness rule	pass if DASM ≥ 0.35 for the family within the span ± 0.5 s; else pass if <i>both</i> listener inventories name the family; if exactly one does, ask the scene check (Section B.1); otherwise drop	DASM_LOCAL_VETO 0.35, DASM_LOCAL_KEEP both, SCENE_FIT_LOGIT
8	inventory keep	a span with confidence ≥ 0.35 for which both inventories exist and neither names its family (or a descendant) is dropped, unless DASM ≥ 0.575 within ± 0.5 s	KEEP_NEEDS_V4_ALL onto, ..._DASM_KEEP
9	band-twin pull	the start moves back to an earlier same-family FlexSED run (≥ 0.5) that ends ≤ 1 s before and starts ≤ 1.5 s before it	BAND_TWIN_PULL 0.5
10	continuation veto	a span is dropped when a same-family FlexSED run (≥ 0.5) began ≥ 1.5 s earlier and is still on at its start (it is a later piece of a sound already shown)	CONTINUATION_VETO 0.5
11	FlexSED veto	a span whose family FlexSED never reaches 0.3 in the clip is dropped	FLEXSED_VETO 0.3
12	PANNs veto	a FlexSED-only span whose family PANNs never reaches 0.05 is dropped unless a listener accepts it	PANNS_VETO 0.05
13	listener band rescue	a FlexSED run with peak in $[0.5, 0.8)$ and no same-family span is added if accepted: peak ≥ 0.6 needs the Qwen inventory; < 0.6 needs both inventories	LISTENER_RESCUE, LISTENER_RULE TIER
14	DASM rescue	a DASM run (≥ 0.575) with no detector span nearby is added if both inventories name the family and the family is new in the clip	DASM_RESCUE, ..._NEW_ONLY
15	onset refinement	for BEATs spans: silence the first t seconds of the first firing window for 26 values of t and take the first t at which 10% of the class evidence is gone; never earlier than the detected start	ONSET_CAM, ONSET_MONOTONE

#	step	rule	variable(s)
16	rescued-span filters	on rescued spans only: FineLAP ≥ 0.329 ; DASM ≥ 0.575 within ± 0.5 s; at most one rescued span per family per clip (the earliest)	FINE LAP_VETO 0.329, LISTENER_DASM_VOTE, LISTENER_ONCE

The DASM thresholds 0.0839 and 0.575 and the FineLAP bar 0.329 were calibrated during development (Appendix E); every other threshold was set once and not tuned on the test set. Steps 4–8 and 12–16 depend on listener answers that are computed from a first pass with the September version of the system (the candidate “pools”), so the final detector cannot run without that pass (Section C.1).

The scene check When exactly one listener names a weak sound, the witness rule asks the vision model whether the sound is plausible in the scene. The span is cut into stretches of about 5 s; for each, six frames from 1 s before to 1 s after are shown with the question “Could the sound of ⟨label⟩ plausibly be heard in this scene? Answer yes or no.” and its negated twin. The answer is read from the first output token as $d = m(Q) - m(\bar{Q})$, where $m = \max$ logit over {yes, Yes} minus $\max$ logit over {no, No}; $d > 0$ counts as yes, and the sound passes if most stretches say yes. Reading the logits replaced an earlier text answer that was cut off after four tokens on 10–15 % of calls (Appendix E).

B.2 What may be drawn

The label filter (`LABEL_FILTER = "depictable", src/labels.py, is_salient_nonspeech`) maps every label to its family and removes: speech with words and its descendants (Speech, Conversation, Narration, Hubbub, Whispering, ...), music and singing and their descendants, the “channel, environment and background” branch, silence and generic labels, textures (Breathing, Rumble, Hum, Whir, Rustle, Wind and its descendants), and labels with no clear source (Sound effect, Onomatopoeia, generic impacts). Everything else is drawable, including non-word vocal events (shout, scream, laughter, cough, snoring). FlexSED and DASM are asked only about the 215 drawable families.

B.3 Scoring harness versus live pipeline

The numbers in this report come from the scoring harness, which runs the pipeline’s own stage-4 functions on cached model scores and replays the gate answers of a full run. A parity check (`benchmark/gold/parity_check.py`) and an independent re-export of every decision (the Decision Inspector, Section C.1) reproduce the published counts exactly (0 of 159 clips differ). The live configuration `use_shipped()` differs from the scored one in these known ways:

- **Kinship direction.** Live: a visible label silences only the same or a more specific label. Scored: either direction. This was a deliberate logic fix made after the scored tables; its effect on the final numbers was not re-measured.
- **Picture end.** Live pictures end at most 1 s after their sound; scored pictures had no cap. Starts – the scored quantity – are identical; on 3 TEST clips a picture ends 0.04–0.38 s earlier live.
- **Subject path.** `main.py` runs planning, the on-screen check and the duplicate check with the scored settings and switches to the final picture settings only for drawing, so the pictures it shows are designed to match the scored run.
- **Pictures.** Scored runs used placeholder or FLUX.1-schnell pictures; the live system uses Qwen-Image with the check. Pictures are not part of the per-sound score.

B.4 Configuration values

Values set by `config.use_shipped()` (`config.py`), read by running it. “ $\rightarrow$ ” gives the module default and the shipped value.

variable	value	meaning
AED_THRESHOLD	0.175	BEATs extraction bar
AED_MIN_DUR	0.5 $\rightarrow$ 0.3	shortest span (s)
DISPLAY_THRESHOLD	0.35	display bar (stage 5)
FLEXSED_BAR	0.8	FlexSED extraction bar
FLEXSED_VETO	0 $\rightarrow$ 0.3	FlexSED clip veto
PANNS_VETO	0.05	PANNS clip veto on FlexSED-only spans
TWIN_MAX	True	twin confidence = stronger side
MIRROR_VETO	0.7	mirror veto bar
LISTENER_CONFIRMED_MIRROR	True	listener can undo the mirror veto
KEEP_NEEDS_V4	True	a keep must also be named in an inventory
MASKED_WEAK_VETO, _AF	True	masked-weak veto
DASM_CLIP_VETO	0.0839	DASM clip veto
DASM_LOCAL_VETO	0.35	witness rule bar
DASM_LOCAL_KEEP	both	both listeners can witness
SCENE_FIT_LOGIT	True	scene check by logit margin
KEEP_NEEDS_V4_ALL	onto	inventory keep incl. descendants
KEEP_NEEDS_V4_ALL_DASM_KEEP	True	DASM can keep it
BAND_TWIN_PULL	0.5	band-twin pull
CONTINUATION_VETO	0.5	continuation veto
LISTENER_RESCUE	True	band rescue
LISTENER_LO	0.4 $\rightarrow$ 0.5	lowest FlexSED peak rescued
LISTENER_RULE	TIER	Qwen alone ≥ 0.6 , both below
DASM_RESCUE, _NEW_ONLY	True	DASM rescue of new families
FINELAP_VETO	0.329	FineLAP veto on rescued spans
LISTENER_DASM_VOTE / _BAR	True / 0.575	DASM vote on rescued spans
LISTENER_ONCE	True	one rescue per family per clip
ONSET_CAM, ONSET_MONOTONE	True	occlusion onsets, never earlier
MAX_SPAN	8.0 $\rightarrow$ None	no length cap
LABEL_FILTER	depictable	what may be drawn
VLM_MODEL	Qwen/Qwen3.8-27B	gate, subject, checks
VLM_THINKING	False	thinking off
VISIBILITY_STRETCH / _RULE	5.0 / majority	gate stretches and vote
KINSHIP_DIRECTED	True (scored: False)	kinship direction
VIDEO_BACKEND	owlv2	stage-2 objects
DEPICT_EVENT	True	depict-event drop
MERGE_GAP	2.0 $\rightarrow$ 2.5	repeat merge (s)
MIN_DWELL	1.5	minimum on screen (s)
MAX_AFTER_END	1.0 (scored: None)	cap past the real end (s)
GROUP_ASK / GROUP_MAX_GAP	True / 8.0	grouping by Qwen3-Omni
MAX_SLOTS	3	simultaneous pictures
GEN_MODEL	Qwen/Qwen-Image-2512	picture model (FLUX retired)
RESOLUTION	1024 $\times$ 1024	picture size
PICTURE_V3, _SCENE, _FINAL, _MAKER	True	subject writing

variable	value	meaning
PICTURE_VERIFY / _TRIES	True / 5	check and redraw
CONFIDENCE_FADE	False	full opacity

C Reproducibility and sources of the numbers

C.1 Reproduction

Code. Release v1.1.0 of the repository. The final system is the configuration `config.use_shipped()` in `config.py`; in the scoring harness it is the variant SHIP8+MD3+WW5+SL (Appendix D).

Environment. Python 3 with PyTorch 2.5.1, transformers 5.16.1, diffusers, faster-whisper, panns-inference, easyocr and sentence-transformers. FineLAP needs transformers 4.51.3 and runs in a second environment, called as a subprocess (variable `FINE LAP_PYTHON`). FlexSED and DASM are imported from their own repositories (paths in `config.py`). Models (Hugging Face identifiers): Qwen/Qwen3.8-27B, Qwen/Qwen3-Omni-30B-A3B-Instruct, nvidia/audio-flamingo-next-hf, AndreasXi/FineLAP (revision b419aa2), CPF2/detect_any_sound, Qwen/Qwen-Image-2512, google/owlv2-base-patch16-ensemble, BEATs iter3+ AS2M cpt2 and PANNs CNN14. Revisions other than FineLAP are not pinned.

Hardware and running time. One GPU with about 80 GB of memory (an NVIDIA H200 was used; two A100 80 GB cards also work). Qwen-Image needs about 57 GB in bf16, and the picture check needs the picture model and the vision model loaded together. Preparing the listener inputs takes about five minutes per clip; each picture try takes about 20 s.

Running the system on one video.

```
python main.py --input clip.mp4 --device cuda
```

This writes `data/output/<clip>_augmented.mp4` (the video with the picture panel) and the intermediate files under `data/work/<clip>/. The run first computes the per-clip model inputs the detector needs (FlexSED, BEATs and PANNs scores, the listener answers, DASM and FineLAP scores), then runs detection, the on-screen check and the pictures. Detection and the on-screen check run with exactly the settings of the scored results; only the picture step uses the final picture settings. A folder of clips is processed by calling main.py once per clip.`

Reproducing the reported numbers. The scoring harness reads the gold labels and the cached model outputs of each clip and replays the pipeline’s own decision functions:

```
python benchmark/gold/final_vs_baselines.py # Table 3: final system and baselines
python benchmark/gold/parity_check.py SHIP8+MD3+WW5+SL # configuration check
python benchmark/gold/inspector_trail_export.py \
  --expect DEV=29/58/15/6/7/2/2.056 --expect TEST=24/65/24/4/15/5/2.409
```

The last command re-runs the final system with every decision logged and stops unless the counts match exactly; it reproduces all 159 clips. The cached model outputs of the benchmark clips are not part of the repository (they are large); they are rebuilt from the clips with `benchmark/gold/tagger_prep.py`.

Decision trail. `docs/inspector2/` is a static page over `data.js` (159 clips) that shows, for every labelled sound and every candidate of the final system, each step it passed or failed, the value, the threshold and the model answers. The logging (`src/trail.py`) only records and never changes a decision.

Where the records are. The development record is in `docs/history/`: pre-registrations with their results (`docs/history/preregistrations/`), review panels, daily notes and analyses; `docs/README.md` gives the map. The gold labels are `benchmark/gold/annotations/gold_AG.json`.

C.2 Sources of the numbers

Every number in the main body comes from one of the files below (paths relative to the repository root). Counts of the final system are also reproduced exactly by the decision-trail export (`docs/inspector2/data_parity.json`, `"exact": true`).

numbers	file
Table 1 (set sizes, needed sounds)	<code>benchmark/gold/annotations/gold_AG.json</code> , counted with <code>benchmark/gold/score_per_sound.py</code>
labelling rules (Section 3.2)	<code>benchmark/gold/tool_template.html</code> , <code>tager/index.html</code> , <code>docs/history/plans/metric_per_sound.md</code>
scoring rules, cost (Section 3)	<code>benchmark/gold/score_per_sound.py</code>
bootstrap and pass rules	<code>benchmark/gold/dev_candidates_check.py</code> ; <code>benchmark/gold/holm_table.py</code>
final system and baselines, Table 3	<code>benchmark/gold/final_vs_baselines.json</code> ; <code>docs/history/review/detector_final_2026-10-02.md</code> ; <code>docs/inspector2/data_parity.json</code>
on-screen weight (Section 6.4)	derived from Table 3; earlier versions in <code>benchmark/gold/visible_weight_sweep.md</code>
fate of misses (Section 6.5)	<code>docs/history/review/inspector_trail_ready.md</code> ; <code>docs/inspector2/data.js</code>
Table 4 (ceiling)	<code>benchmark/gold/ceiling_ship7.md</code>
September gate comparison (Appendix E)	<code>benchmark/gold/holm_test_final_v33_test_bench.json</code>
gate accuracy, failure modes	<code>docs/history/review/vlm_panel_2026-10-01/f1_audit.md</code>
listener agreement on the screening set	<code>docs/history/review/improvements_log_2026-09-30.md</code> ; <code>docs/history/preregistrations/prereg_round13_detector_push.md</code>
onset timing	<code>benchmark/eval_dcase_onset.json</code>
DCASE visibility check	<code>benchmark/eval_dcase_visibility.json</code>
picture glance test	<code>docs/history/preregistrations/picture_v3_prereg.md</code>
picture check validation	<code>docs/history/preregistrations/freeze_picture_setup_2026-09-25.md</code>
label re-checks, missing-sound checks	<code>docs/history/preregistrations/prereg_round13_detector_push.md</code>
test-set reads (Section 9)	<code>docs/history/preregistrations/prereg_round13_detector_push.md</code> ; <code>docs/history/earlier_drafts/thesis/ch5_results.md</code>
configuration (Appendix B)	<code>config.py</code> (<code>use_shipped()</code>)
development history (Appendix E)	<code>docs/history/</code> (preregistrations, panels, daily notes, review)

D Glossary of internal names

The development logs in `docs/history/` and the scoring code use short codes. The main body of this report does not; this table maps each code to the plain name used in the report.

code	plain name / meaning
<i>Systems and versions</i>	
D', SHIP8+MD3+WW5+SL	the final system (release v1.1.0, <code>config.use_shipped()</code>)
D	the final system with the scene check read as text
B, SHIP8+MD3	the version before the last three rules (witness rule, scene check, logit reading)
B0, B0r	the September baseline (<code>config.use_scored()</code>); B0r = B0 re-run in the same job as a candidate
B1	the September baseline with a BEATs self-veto (rejected)
SHIP1–SHIP8	successive accepted versions (Appendix E)
v1, v2, v3, v4	earlier versions of the system (August–September; Appendix E)
<i>Data</i>	
DEV-49, TEST-60	the older parts of the development and test sets
DEV2, TEST2	the newer parts (tagger clips)
merged DEV / TEST	the development set (71 clips) / the test set (88 clips)
the 415, held-out slice B	the screening set (415 AudioSet-Strong clips) 30 AudioSet-Strong clips with a masked sound, reported separately
<i>Components</i>	
TO1+F7F8, TIER	listener rescue of weak FlexSED detections (Qwen alone for strong runs, both listeners for weak)
V4	a listener’s open list of heard sounds (“inventory”)
N2b	masked-weak veto
DR2	DASM rescue of new families
K-V4	a listener keep must also be named in an inventory
DV	DASM clip veto
BTP	band-twin pull (start a late picture at an earlier weak run)
CONT	continuation veto
FLAP	FineLAP veto on rescued sounds
K4AO, K4A-D	inventory keep (drop a picture neither listener lists; K4A-D: unless DASM hears it)
MD3	shortest detected sound 0.5 → 0.3 s
GRP, GROUP	same-family grouping by Qwen3-Omni
WW, WW5	witness rule; WW5 = witness rule with the scene check
SL, 60L	scene check read from the yes/no logits
DEPICT	depict-event drop
EXPECT	“expected sound” rules: name sounds expected off screen, then confirm (rejected)
<i>Process</i>	
Round n , R n	a numbered, pre-registered development round
amendment n	a numbered change to the September pre-registration (<code>prereg_v4.md</code>)

E Development history

This appendix is the record of how the final system was reached: how the data grew, the working rules, the earlier versions of the system, every accepted change, the main rejected ideas, the history of the pictures and of the evaluation, and a round-by-round table. Internal codes from the development logs appear here in parentheses so that each row can be found in `docs/history/`; Appendix D maps them to plain names.

E.1 How the data grew

Clip-level benchmark (August–September). 274 short clips (10–20s) were labelled by the author into four scenarios: unseen ambient (48 clips), mixed (25), seen ambient (132) and no ambient (69). Four sourcing strategies (keyword scraping, AudioSet strong segments, curated film scenes, UnAV-100 overlap moments) and an automatic filter all landed at 15–20 % usable clips, which is why the set stopped at 274 rather than the planned 300. A test set of 100 clips (25 per scenario) was used by the early systems and the model judge (Appendix E.7).

Per-sound labels (22 September). After the judge-based evaluation was abandoned, clips were labelled per sound. The first export had 139 clips: 109 benchmark clips and 30 AudioSet-Strong clips with a consequential sound at least half under speech or music (“slice B”, always reported separately). The 109 were split by provenance into 49 development clips (“DEV-49”, clips already touched by earlier runs) and 60 test clips (“TEST-60”, labelled after the system was frozen). The September test table (Section E.9) is on TEST-60.

New clips (29 September – 1 October). 50 more clips were labelled with a stand-alone tool in three batches: 31 YouTube clips found by scene type and 19 AudioSet-Strong evaluation clips. They were split into 22 development (“DEV2”) and 28 test (“TEST2”) clips by a hash of the clip name or by balanced pairs, and merged on 30 Sept into the development set ($49 + 22 = 71$ clips) and the test set ($60 + 28 = 88$ clips) used in the body.

Table 9: Parts of the two evaluation sets.

part	clips	sounds	needed	imp. 3	clips w. needed	unseen/mixed/seen/none
DEV-49	49	95	38	14	24	16/9/16/8
DEV2	22	59	20	3	14	8/7/7/0
TEST-60	60	94	43	11	28	13/15/12/20
TEST2	28	60	22	2	15	8/8/10/2

Needed-sound count. The development count moved during the last week: 55 at the first merged scoring, 58 after the visibility re-check (three sounds changed from seen to needed), 57 after a control round, and back to 58 after a correction. Every system was re-scored on each version; the test labels were never changed.

AudioSet sets. A calibration set of 280 AudioSet-Strong clips decided detector rounds 4–12 with a cost on timed AudioSet labels. Round 12 showed that under a corrected version of that cost the detector of the time was worse than showing nothing (1.650 vs 0.986 per clip), because AudioSet labels only a few sounds per clip; the set was retired as a decision test. The screening set of 415 clips was then used only for single audio rules, and a fresh set of 422 clips built for a last confirmation was never scored.

E.2 Working rules

From 13 Sept the following rules governed every change (`docs/history/plans/plan_robustness.md`, `docs/history/preregistrations/prereg_round13_detector_push.md`):

1. **Pre-register.** Before a run, write the idea, the exact rule, the data it is decided on and the pass bar, and commit it. Results are appended below the pre-registration, never edited into it.
2. **Decide on development data only.** The development set (or, for audio-only rules, the held-out 415) decides. The test set is read after a change ships and is only reported. The

rule was re-affirmed on 1 October after one test-informed choice was identified and reverted (Section 9).

3. **General rules only.** A rule must make sense for any video, run on one video at a time, and not name a clip or a single sound class (a rule that only ever fires on bells was rejected for this reason, Round 64).
4. **Compare with the shipped pipeline.** Every candidate is scored as a full-pipeline variant on the same clips as the current shipped system, not on a proxy; a component screen may stop an idea early but cannot ship it.
5. **Hits first.** Losing a few hits is acceptable only if many wrong pictures go (the fewer-pictures clause, Section 3.5).
6. **Consult and record.** Major choices were reviewed by panels of independent model reviewers (Claude, with assigned roles, two to five per panel, one to four rounds), whose reports are in docs/history/review/ and docs/panel_*.md; every decision and its reason is in a dated log.

E.3 Phases

Table 10: Project phases. Scores in each row are in that phase’s own metric and are not comparable across rows.

dates	system	scored by	outcome and why it changed
6 Aug	proposal: 7 stages (audio, VLM, ASR, BEATs/PANNs, reasoning, FLUX/SDXL, evaluation)	planned: VLM describes, LLM writes reference, LLM judges	pivot from an earlier speech-illustration project
7 Aug	v1: PANNs (bar 0.15) → Openverse image retrieval → side panel; no gate	demo clips	needed a “not seen” decision
7 Aug–7 Sep	CLIP zero-shot visibility gate; clip-level benchmark (274 clips)	clip-level gate accuracy	CLIP gate 49.3%: 71 of 103 seen clips still augmented – “object presence is not event visibility”
08–11 Sep	PANNs + OWLv2 + per-sound VLM gate (Qwen2.5-VL-7B, “name it, don’t confirm”); SDXL	100 clips × 3 systems, Mistral-7B judge 0–4, model-written reference	blind 2.94 > caption 2.68 > gated 2.41; retrieval beat SDXL by +0.17/+0.18; reference later found circular
12–17 Sep	v3: BEATs (replaced PANNs after 11 of 14 complaints traced to its labels), hysteresis + occlusion onsets, 7B gate with 3 votes, FLUX.1-schnell	human-grounded judge reference, dev 174 / test 100	tie 3.17 vs 3.16; gate wins where nothing is due (+0.26), loses where a picture is due (−0.24)

dates	system	scored by	outcome and why it changed
18–21 Sep	v4: one state-of-the-art swap per stage; Qwen3.8-27B gate; FLAM and PretrainedSED detectors tried; rubric-enforced judge	judge with a cap on redundant pictures	PretrainedSED rejected three times; 13 silent bugs found (amendment 3), all earlier judge rows kept as history only
22–23 Sep	per-sound gold (139 clips); BEATs $\cup$ FlexSED (bar 0.8), FlexSED veto 0.3, PANNs veto 0.05	per-sound F1, then viewer cost (amendment 9)	F1 difference null (+0.028); wrong pictures and precision significant
24–27 Sep	monotone onsets, no length cap; final September test table (amendment 21); Qwen-Image frozen as picture setup	Holm-corrected table on TEST-60	“Holm wins, F1 null” (Section E.9)
27 Sep	detector rounds 1–3: cascade, PE-A-Frame, Qwen3-Omni as verifier	DEV-49 + AudioSet 280 / 415	none passed; detection declared closed, reopened next day
28 Sep	rounds 4–12 on AudioSet sets: EAT, Dasheng, DASM, SAM-Audio, 17 ideas, local contrast, DASM rescue, masker tests	cost on AudioSet labels	all failed on the 415; round 12 showed the AudioSet cost cannot track the task: harness retired
28–30 Sep	rounds 13–14: listener rescue (Qwen3-Omni, Audio Flamingo Next) + DASM vote; tagger clips added	DEV-49, then merged DEV	first gains in hits (TO1+F7F8)
30 Sep–01 Oct	ship chain SHIP1–SHIP8 on merged DEV/TEST	merged DEV, pass rules	DEV 18/51/3.690 $\rightarrow$ 28/21/2.282 (Table 11)
01–02 Oct	rounds 46–66: display rules, witness rule, scene check, many gate ideas	merged DEV, held-out 415	the final system: DEV 29/15/2.056; frozen on 2 Oct

E.4 The shipped chain

Table 11 is the backbone of the history: every change that entered the final system, in order, with its development and test numbers on the merged sets. Development numbers use the corrected gold (58 needed sounds); test numbers are the reads made after each change shipped. Sources: `docs/history/review/improvements_log_2026-09-30.md`, `docs/history/review/detector_final_2026-10-02.md`, `docs/history/preregistrations/prereg_round13_detector_push.md`.

Table 11: Changes that shipped, from the old pipeline to the final system. Hits / wrong / cost; DEV hits out of 58, TEST out of 65. The p in the TEST column is one-sided, against the old pipeline B0r for SHIP5–SHIP8 and against B for D and the final system.

step (name in logs)	what it adds	merged DEV	merged TEST
old pipeline (B0r)	September test-table system: BEATs $\cup$ FlexSED, FlexSED and PANNs vetoes, Qwen3.8 gate	18 / 51 / 3.690	21 / 40 / 2.909
TO1+F7F8 (R14)	listener rescue of weak FlexSED runs (tiered: Qwen alone above 0.6, both listeners below), one rescue per family, mirror veto with listener keep, DASM vote on rescues	26 / 45 / 3.070	22 / 38 / 2.818
+ N2b, DR2, K-V4 (R16–21)	masked-weak veto; DASM rescue of new families; a listener keep must also be named in an inventory	27 / 38 / 2.817	23 / 36 / 2.727
+ DV (R28)	drop a sound DASM never hears in the clip	27 / 31 / 2.620	23 / 34 / 2.682
+ BTP (R30)	start a late picture at an earlier weak run of the same sound	28 / 30 / 2.535	23 / 33 / 2.659 (p 0.033)
+ CONT (R31)	drop a picture that is only a later piece of a sound already going on	28 / 25 / 2.394	24 / 32 / 2.591 (p 0.025)
+ FineLAP veto (R31)	a 2025 frame-level model double-checks rescued sounds	28 / 24 / 2.366	24 / 31 / 2.568 (p 0.015)
+ K4A-D = SHIP8 (R35)	drop a picture neither listener names, unless DASM clearly hears it	28 / 21 / 2.282	23 / 29 / 2.568 (p 0.017)
+ merge gap 2.5 s	continuing sounds with short breaks no longer drew a second picture	28 / 20 / 2.254	23 / 28 / 2.545
+ grouping ≤ 8 s (R47)	Qwen3-Omni: same continuing sound or a new event?	28 / 18 / 2.197	22 / 26 / 2.545
+ shortest sound 0.3 s (R48)	short real sounds (a laugh burst) were thrown away	29 / 18 / 2.141	22 / 26 / 2.545
+ depict-event (R57)	drop a picture whose event is visibly happening	29 / 17 $\rightarrow$ 18 ^a / 2.141	22 / 26 / 2.545
= B (SHIP8+MD3)	base of the last rounds	29 / 18 / 2.141	22 / 26 / 2.545
+ witness rule + scene check (R53+R60) = D	weak sounds need DASM, both listeners, or one listener and a plausible scene	29 / 14 / 2.028 ^b	24 / 23 / 2.386
+ logit readout (R60L) = final system	scene check read as a probability, no cut-off answers	29 / 15 / 2.056	24 / 24 / 2.409 (p 0.034 vs B)

^a Round 57 scored 29/17 with 26 listener answers missing from a cache; after the cache was filled (1 Oct evening) the same system scores 29/18/2.141, so its net development effect is zero. ^b D’s 2.028 contains one gain caused by a cut-off answer (“based on the”) that removed a wrong *Glass* picture; the logit readout of the final system cannot be cut off, so that picture returns (+1 wrong). The pass bar of Round 60L was set against B, not against D, for this reason.

What the table shows. From the old pipeline to the final system the development set gained 11 hits and lost 36 wrong pictures (cost 3.690 $\rightarrow$ 2.056). The test set gained 3 hits and lost 16 wrong pictures (2.909 $\rightarrow$ 2.409). About a third of the development gain in cost carried over to the test set; with well over a hundred development variants tried, some of the development gain is expected

to be selection bias (the winner’s curse), and the test numbers are the better estimate of what a new video would see.

E.5 What was tried and rejected

Most ideas failed. Table 12 groups the main ones by the question they tried to answer, with the reason from their own pre-registered test. All were decided on the development set or the held-out 415.

Table 12: Main rejected ideas (selection). Hits / wrong / cost on merged DEV unless stated.

idea (round)	what it did	why it failed
<i>Hear more sounds</i>		
other detectors (Sep, rounds 1–8)	FLAM, PretrainedSED (5 backbones), AST, CED, SSLAM, EAT, Dasheng, PE-A-Frame, CLAP verifier, tagger ensembles	recall rose only with many more false spans; PretrainedSED lost three times; ensembles mis-scaled
source separation (Sep, R7)	Demucs, SAM-Audio in front of BEATs	separated audio is out of the detectors’ training distribution: recall under speech fell 9.5% $\rightarrow$ 4.8%; SAM-Audio failed its quality check (music never removed)
listener as verifier (R13)	ask Qwen3-Omni yes/no, multiple choice, open list on every weak candidate	finds 5–8 dropped real sounds but adds about 3 wrong pictures per hit
newer listeners (R31–34)	MOSS-Audio, SpotSound, Kimi, Step-Audio	3–10 wrong per extra hit; SpotSound answered yes to every query
EXPECT chain (R40–42b)	the scene VLM or a listener names sounds expected off screen; FineLAP times them; DASM confirms	best DEV 31/26/2.254 (+3 hits), but TEST 25/35/2.614 (worse); the gate cannot refuse a wrong off-screen name
short twins, band lists (R49, R56)	add weak or short detections both models half-hear	BANDLIST 0 hits, +6 wrong; TWIN-SHORT 27/36/2.761
tagger ensemble (R63)	calibrated EAT/SSLAM ensemble on BEATs windows	on the 415: 33% fewer spans but recall 0.152 vs 0.197
<i>Remove wrong pictures</i>		
stricter witnesses (R53 variants)	require DASM or both listeners for every weak sound	27/13/2.113: –5 wrong but 2 real hits lost (fails the 3:1 clause)
named veto, relabel gate (R39–41)	drop or rename a picture when the gate or the listeners name a different sound	real hits lost for no cost gain
held-out family prior (R41)	drop families a detector gets wrong > 70% of the time on the 415	every affected picture had been heard by a listener: no change
<i>Better on-screen decision</i>		
other gate signals (Sep, am. 14–20)	OWLv2 or SAM 3 vote, video input, synchrony, “any stretch” rule	each removed exactly as many needed sounds as wrong pictures it saved (break-even at $\beta = 2$)
SSL-SaN, SigLIP, BOX (R30–38)	audio-visual localiser; picture-vs-frame similarity; VLM draws a box on the source and checks the crop	each un-silenced real visible sounds; BOX-2 in the full pipeline 29/26/2.366
synchrony (R38)	Synchformer: does on-screen motion move with the sound?	as a veto, +3 needed kept but 5 visible calls lost; as an “add seen” vote, on the bar but changed no shipped picture

idea	what it did	why it failed
human-style chain (HUMAN, R50, R66)	name the source, find it, zoom, ask “is it making the sound now?”	the VLM names “nothing” on most stretches; NAME-ALL passed its gate screen but lost 3 real hits in the pipeline (26/12/2.141)
sign/event and logit gate (R54, R62)	ask for visible effects; read the gate as a logit margin	letter-position bias or “no” habit; best AUROC 0.647, below the 0.65 bar
look-alike reasoning (R61)	“no train here, so it is the washing machine”	works on the laundromat, but also explains away real sounds (a crying sound “explained” by a visible parrot)
bell rule (R64)	concealed actions (a bell in a tower) count as not visible	passed DEV (30/14/1.972) but fires only on bells: not general, not shipped
<i>Timing and grouping</i>		
change-point onsets (R33)	SEBB onset post-processing	7 hits lost
earliest onset (R38), repeat rules (R31, R35)	start at the first detector to hear a sound; require silence before a repeat	hits lost or no net gain
pause questions (R51, R55)	ask whether a 2-s gap is a pause or a new sound	neither the listener nor the detectors can tell a pause from a dropout
context ± 5 s (R59)	give the listener more context	it names what it <i>sees</i> , not what it hears

The main lesson. Two patterns recur in this table. First, the missing sounds can often be found – a listener asked about the whole clip names the hammer, the train, the bell – but every way of adding them brings two to four wrong pictures per hit, because nothing downstream can tell a right off-screen name from a wrong one. Second, every change to the gate moves along the same trade-off line: it silences more on-screen sounds only by silencing about as many needed ones. Section 8 measures both.

E.6 Pictures

The picture model changed four times: Openverse retrieval (v1), SDXL and SDXL-Turbo, FLUX.1-schnell (chosen over PixArt- Σ on twelve real prompts, 13 Sept), and Qwen-Image-2512 (frozen 25 Sept, shipped 28 Sept). On 9 Sept a retrieval-versus-SDXL ablation on the same 100 clips favoured retrieval by +0.17 (gated) and +0.18 (blind) judge points, but 55 % of retrieved images carried a non-commercial or no-derivatives licence and 595 uses drew on only 60 distinct images, so generation was kept. The decisive picture test was a blind *glance test* by the author on 25 Sept: 54 sounds from 50 never-annotated clips, each picture shown for 1.5 s at 384 px with no name, and the author typed what the sound was. FLUX was recognised 14/54 times; Qwen-Image with the same text 26/54 (+0.222 [+0.074, +0.370]); Qwen-Image with a richer subject text 32/54, but that text invented an object (a thud drawn as a door) and was rejected. One of the 26 Qwen-Image pictures was later found to send a false message (a telephone bell drawn as a desk bell), so the +22-point result is reported as “not clean”. A planned sealed confirmation sitting was cancelled on 28 Sept and replaced by the check-and-redraw loop of Section 5.5. A wording test the same day kept the hand-written subject table: 7 of 7 known cases right, against 3 of 7 for two generic VLM wording methods (docs/history/analyses/picture_sense_test_2026-09-28.md).

E.7 Earlier evaluation by model judge (superseded)

The proposal planned an automatic evaluation: a VLM describes the augmented video, an LLM writes a reference of what the viewer should learn, and an independent LLM scores the description against the reference on a 0–4 scale. It was built and run, and it failed in instructive ways:

1. **A reference written by the system’s own models is circular.** The first reference was written by the system’s VLM from the system’s detections; it rewarded whichever system repeated the detector (blind 2.94 > caption 2.68 > gated 2.41 on 9 Sept). A second judge of another family agreed with the first (quadratic $\kappa = 0.753$), which shows only that two judges can share a bias.
2. **A reference written by independent models cannot decide visibility.** A reference built by an audio LLM, CLAP, a different VLM and a different LLM agreed with the human “nothing is missing” label on 55 of 100 clips – chance. An audio-visual model asked with and without the soundtrack reached 49.5 % balanced accuracy (bar 67.7 %); a “gap-closing” score had AUROC 0.529 (bar 0.75).
3. **The rubric was asymmetric.** A withheld picture scored 0 where one was due, a redundant one about 3 where none was due, so the rubric punished staying silent – the very behaviour the gate exists for. A capped rubric (19 Sept) and a human-grounded reference fixed the direction but not the noise.
4. **Judges cannot tell a picture from a text tag.** The best judge (Gemma-4-31B seeing the pictures directly, passing all three trust checks against the author’s labels) scored pictures and text tags alike ($-0.04 [-0.22, +0.14]$), because it reads a tag as the label. Six automatic picture checkers also failed calibration against the author (for example $\kappa = 0.21$ to 0.45 against bars of 0.5).

A five-reviewer scoring panel on 19 Sept moved the primary score to per-sound matching against human labels (Section 3) and kept the judge as a secondary instrument. The answer to RQ4 is built on this history.

E.8 How the cost changed during development

Figure 9 shows the cost after each accepted change, from the system of the September test table (“September version”) to the final system. The development cost fell by 1.634 per clip and the test cost by 0.500; the gap is expected, because changes were chosen on the development set.

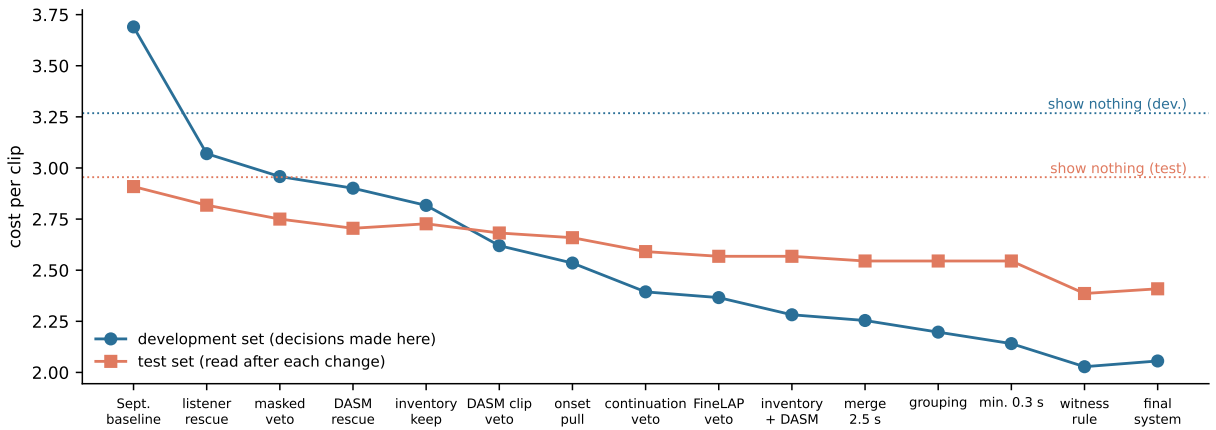


Figure 9: Cost per clip after each accepted change. Decisions were made on the development set; the test set was read after each change and only reported. Dotted lines: showing nothing.

E.9 The September comparison of the on-screen check

Before the final detector existed, the on-screen check was compared with and without it on the 60 older test clips (same gate model and rule, the September detector). Two-sided p -values with Holm correction:

Measure	With minus without	95 % interval	Holm p
Wrong pictures per clip	-0.517	[-0.800, -0.283]	< 0.001
Viewer cost per clip	-0.833		0.012
Precision	+0.137		0.010
Clips handled perfectly	+0.219		0.006
F1	+0.059	[-0.030, +0.144]	0.18 (not significant)

The same pattern holds for the final system (Table 3): the check removes many wrong pictures at the price of a few hits, which F1, weighing both equally, does not reward.

E.10 Comparisons between versions

The final development phase compared each accepted change with the version before it and, early on, with the September baseline (“B0r”). Paired clip bootstrap on the test set, 2000 draws; d = mean per-clip cost difference; one-sided p (a two-sided value is about twice as large).

comparison	d	95 % interval	one-sided p
final system vs the version before its last three rules (B)	-0.136	[-0.295, 0.000]	0.034
same, scene check read as text (D) vs B	-0.159	[-0.341, -0.023]	0.0145
intermediate version (SHIP8) vs September baseline	-0.341	[-0.636, -0.045]	0.017
first listener stack (TO1+F7F8) vs September baseline, pre-registered	-0.091	[-0.250, +0.045]	0.132

Version B scored 29 / 18 / 2.141 (development) and 22 / 26 / 2.545 (test); D scored 29 / 14 / 2.028 and 24 / 23 / 2.386. The four early test reads of 30 Sept were also corrected for multiplicity (Holm); none was significant after correction.

Repeat-merge gap. Re-reading the same saved pictures at different merge gaps moves wrong-picture counts but not the ranking (hits / wrong / cost):

gap (s)	SHIP8, dev.	baseline, dev.	SHIP8, test	baseline, test
1.5	28 / 23 / 2.338	18 / 57 / 3.859	23 / 30 / 2.591	21 / 44 / 3.000
2.0	28 / 21 / 2.282	18 / 51 / 3.690	23 / 29 / 2.568	21 / 40 / 2.909
3.0	28 / 18 / 2.197	17 / 43 / 3.521	22 / 27 / 2.568	20 / 33 / 2.795

The baseline gains more from a longer gap (many of its pictures are repeats), so the final system (gap 2.5s) against the baseline (gap 2.0s) is not a like-for-like comparison.

E.11 Round-by-round record

One row per round or main arm, in order. Results are hits / wrong / cost on merged DEV (55 or 58 needed, see Appendix E.1) unless marked; “415” = held-out AudioSet-Strong screen; “old DEV” = DEV-49. Full text, bars and line numbers: docs/history/preregistrations/prereg_v4.md (amendments), docs/prereg_round4..12*.md, docs/history/preregistrations/prereg_round13_detector_push.md (rounds 13–66). Counted from that file: rounds 13–28 ran 129 new full-pipeline variants (106 on DEV-49 alone), rounds 29–41 ran 21 more and 42 component screens, rounds 42–57 ran 5 full variants and about 30 screens, rounds 58–66 four full variants and 15 screens.

round	idea	result	verdict
<i>September amendments (prereg_v4)</i>			
am. 5	per-sound gold, F1 headline	Δ F1 +0.028 [-0.022, +0.074]; precision, false alarms significant	F1 null

round	idea	result	verdict
am. 8	FlexSED union, bar 0.8	passes 3/3 rules	shipped
am. 9	viewer cost, $\beta = 2$ (post hoc)	gated 3.36 vs blind 4.24; silence 2.79	reported metric
am. 10	FlexSED veto 0.3	4.12 $\rightarrow$ 3.59	shipped
am. 11–15	per-family bars; speech-removed view; family silence; any-stretch rule; OWLv2 vote	each worse or neutral	rejected
am. 16	PANNs veto 0.05	3.02 / 3.10	shipped
am. 17	tagger ensemble	CED mis-scaled	rejected
am. 18–20	visibility signals (video, SAM 3, synchrony)	all at break-even 2:1	rejected
am. 21	final September TEST table	Table E.9	reported
am. 22–25	cascade, PE-A-Frame, Qwen3-Omni verifier	17 hits, cost 4.82; AUROC 0.53; AUROC 0.661	rejected
<i>Rounds 4–12 (AudioSet harness, 28 Sept)</i>			
4	BEATs self-veto replaces PANNs veto	415 neutral; later worse on DEV and TEST (2.83 vs 2.63, p 0.043)	reverted 30 Sept
5	EAT, Dasheng	415 +0.087	fail
6	DASM as detector	415 +0.294 [+0.111, +0.511]	fail (kept as witness)
7, 7b	SAM-Audio separation	quality check failed	stop
8	17 ideas (I1–I10)	closest 17 -0.039 , p 0.061	fail
9	local contrast J2	415 pass, DEV 14 $\rightarrow$ 13 hits	not shipped
10	DASM rescue R0–R7	9 of 61 targets reachable	hold
11	masker tests	all worse	no pick
12	cost v2 on AudioSet	shipped 1.650 vs nothing 0.986	harness retired
<i>Rounds 13–28 (29–30 Sept)</i>			
13	listener rescue, many question forms; R13-1 twin-max	~ 3 wrong per extra hit; R13-1 TEST = B0r	R13-1 kept inside TO1
14	TO1+F7F8	old DEV 18/24/2.45 vs 2.78; merged 23/48/3.155; TEST 22/38/2.818 (p 0.13)	shipped (SHIP1)
16	N2b masked-weak veto	23/44/3.042; TEST 22/35/2.750	shipped
19–20	DR2 DASM rescue	24/44/2.986	shipped (SHIP2)
21	K-V4 inventory keep	24/41/2.901	shipped (SHIP3)
22–27	N2c, stacks, K2 variants	N2c 22/38/2.930 lost a hit	rejected
28	DV DASM clip veto	24/34/2.704; TEST 23/34/2.682	shipped (SHIP4)
<i>Rounds 29–41 (30 Sept – 1 Oct)</i>			
29	F8-U, DV-L, DV-G; V4D; GA	+2 hits for +7 wrong; no-op; -5 hits; STOP; STOP	rejected
30	BTP band-twin pull	25/33/2.620; TEST 23/33/2.659	shipped (SHIP5)
30	K4AO; PIC-SIM; RT	held; nothing flips; no-op	held / rejected
31	CONT continuation veto	25/28/2.479; TEST 24/32/2.591	shipped (SHIP6)
31	FineLAP veto 0.329	25/27/2.451; TEST 24/31/2.568	shipped (SHIP7)
31	RPT-S, PMC, N2c, DVG, PTC, SUBJ, OM, SPOT, H2	screens or arms without gain	rejected
32	FineLAP in DASM’s seat; DV2	+11 wrong; Thunder lost	rejected
33	SPOT-V, change-point onsets, K4A-4E, SSL-SaN gate	all “Yes”; -7 hits; no-op; STOP	rejected
34	MOSS-Audio as listener	+4/ -9 needed, +63 other	rejected
35	DBX, DBR	premise wrong; cost not lower	rejected
re-check	visibility re-check (gold)	3/14 flips; control 0/20	gold change
35	K4A-D (K4AO + DASM keep)	28/58, 21/2.282; TEST 23/29/2.568 (p 0.017)	shipped (SHIP8)
36–37	DASM impact peaks + VLM naming; BOX gate	+2 cross; -7 seen silenced	rejected
38	CF, BOX-2, GBTP, SHIP8+BOX2, SYNC, SYNC-2	best BOX-2 arm 29/26/2.366	rejected
39	DETACHED-ADD, CONTRAST, MAKER-VIS, RELABEL-GATE	+7 wrong; +142 other; -9 hits; hit lost	rejected
40–40e	EXPECT chain	A4: DEV 31/26/2.254, TEST 25/35/2.614	not shipped
41	AGREE, NAMED-VETO, PRIOR415	below bar; hits lost; no change	rejected
<i>Rounds 42–57 (1 Oct)</i>			
42	held-out check of EXPECT-A4 audio chain	74/92 = 0.80 precise vs 0.38–0.39 raw	diagnostic
42b	EXPECT-A5 (7 vouched families)	DEV 30/21/2.169; TEST 23/31/2.614	not shipped

round	idea	result	verdict
43–43c, 50	gate gold-set variants (HUMAN, HUMAN-2, HUMAN-ADD)	one sound short of the bar at best	rejected
44	AVNAME (eyes + ears name the sound)	rename 15/34; drop 27/17/2.225, TEST no change	rejected
45	AGREE-EARS (second listener family)	DEV 31/24/2.197; 415 0.833 < 0.85; TEST 2.659	rejected
46, 46b	gold scope (blind rating of wrong pictures)	≈ 1 in 6 is a real unlisted sound	no gold edit
–	merge gap 2.5 s	28/20/2.254	shipped
47	GRP-A grouping by Qwen3-Omni, ≤ 8 s	28/18/2.197	shipped
48	MD3 shortest span 0.3 s	29/18/2.141	shipped (= B)
49	BANDLIST	0 hits, +6 wrong	rejected
51, 52b, 55	pause rules, kin-keep, frame-pause	415 STOP	rejected
52	SK7 strong keep	29/22/2.254	rejected
53, 53d	WEAK-WITNESS alone	27/13/2.113 (2 hits lost); 27/17/2.225	fail alone
53b, 53c	witness bars from the 415	outside allowed range	stop
54, 54b	SIGN (visible effect)	0 flips; “no” on 9/10	rejected
56	TWIN-SHORT	27/36/2.761	rejected
57	DEPICT-EVENT	29/17 (net 0 after cache fill)	shipped
<i>Rounds 58–66 (1–2 Oct)</i>			
58, 58b	tight listener cut; FlexSED witness	415 precision 0.400, 0.297	stop
59	context ± 5 s	415 0.333	stop
60	SCENE-MARGIN + witness (= D)	29/14/2.028; TEST 24/23/2.386 (p 0.0145)	shipped
61b–d	look-alike reasoning (Qwen, Gemma)	hits lost in every variant	fail
62, 62b	logit gate; PRIOR	AUROC 0.647 / 0.644 < 0.65	stop
60L	scene check by logit margin (= the final system)	29/15/2.056; TEST 24/24/2.409 (p 0.034)	shipped, frozen
63	tagger ensemble	415 recall 0.152 vs 0.197	stop
64	concealed action {Bell}	30/14/1.972 (one clip)	not general, not shipped
65	perceptual-onset return	415 0.115	stop
66	NAME-ALL crop margin	gate screen passed; DEV 26/12/2.141 (3 hits lost)	fail

References

- [1] O. Alonzo, H. V. Shin, D. Li. Beyond subtitles: Captioning and visualizing non-speech sounds to improve accessibility of user-generated videos. *ASSETS*, 2022. doi:10.1145/3517428.3544808.
- [2] K. Alhamoud et al. Vision-language models do not understand negation (NegBench). *CVPR*, 2025. arXiv:2501.09425.
- [3] Qwen Team (S. Bai et al.). Qwen2.5-VL technical report. arXiv:2502.13923, 2025.
- [4] Ç. Bilen, G. Ferroni, F. Tuveri, J. Azcarreta, S. Krstulović. A framework for the robust evaluation of sound event detection. *ICASSP*, 2020. arXiv:1910.08440.
- [5] B. C. Biner, F. M. Sofian, U. B. Karakaş, D. Ceylan, E. Erdem, A. Erdem. SonicDiffusion: Audio-driven image generation and editing with pretrained diffusion models. arXiv:2405.00878, 2024.
- [6] H. Chen, W. Xie, A. Vedaldi, A. Zisserman. VGGSound: A large-scale audio-visual dataset. *ICASSP*, 2020. arXiv:2004.14368.
- [7] S. Chen, Y. Wu, C. Wang, S. Liu, D. Tompkins, Z. Chen, F. Wei. BEATs: Audio pre-training with acoustic tokenizers. *ICML*, 2023. arXiv:2212.09058.
- [8] Y. Chu et al. Qwen2-Audio technical report. arXiv:2407.10759, 2024.
- [9] Described and Captioned Media Program. Captioning Key: Sound effects and music. <https://dcmp.org/learn/captioningkey>, accessed 27 September 2026.

- [10] C. de Lacerda Patata, S. Hassan, N. Tinker, R. L. Peiris, M. Huenerfauth. Caption Royale: Exploring the design space of affective captions from the perspective of deaf and hard-of-hearing individuals. *CHI*, 2024. doi:10.1145/3613904.3642258.
- [11] C. Drummond, R. C. Holte. Cost curves: An improved method for visualizing classifier performance. *Machine Learning* 65(1):95–130, 2006.
- [12] B. Elizalde, S. Deshmukh, M. Al Ismail, H. Wang. CLAP: Learning audio concepts from natural language supervision. *ICASSP*, 2023.
- [13] D. Ge, S. Liu, C. Gong, X.-L. Zhang, C. Zhang, X. Li. Towards expressive and faithful audio-to-image generation: A unified multimodal dataset and synthesis framework (AudioCanvas). *ACM MM*, 2026. arXiv:2608.09529.
- [14] A. Gavriely. Visual Augmentation of Audio Semantics for Accessibility: code, labels and report, release v1.1.0. GitHub, 2026. <https://github.com/adamgavriely/MscFinalProject/releases/tag/v1.1.0>.
- [15] J. F. Gemmeke, D. P. W. Ellis, D. Freedman, A. Jansen, W. Lawrence, R. C. Moore, M. Plakal, M. Ritter. Audio Set: An ontology and human-labeled dataset for audio events. *ICASSP*, 2017.
- [16] T. Geng, T. Wang, J. Duan, R. Cong, F. Zheng. Dense-localizing audio-visual events in untrimmed videos: A large-scale benchmark and baseline. *CVPR*, 2023. arXiv:2303.12930.
- [17] Y. Gong, Y.-A. Chung, J. Glass. AST: Audio spectrogram transformer. *Interspeech*, 2021. arXiv:2104.01778.
- [18] J. Hai, H. Wang, W. Guo, M. Elhilali. FlexSED: Towards open-vocabulary sound event detection. *WASPAA*, 2025. arXiv:2509.18606.
- [19] D. Jain, K. Mack, A. Amrous, M. Wright, S. Goodman, L. Findlater, J. E. Froehlich. HomeSound: An iterative field deployment of an in-home sound awareness system for deaf or hard of hearing users. *CHI*, 2020. doi:10.1145/3313831.3376758.
- [20] D. Jain, L. Findlater, J. Gilkeson, B. Holland, R. Duraiswami, D. Zotkin, C. Vogler, J. E. Froehlich. Head-mounted display visualizations to support sound awareness for the deaf and hard of hearing. *CHI*, 2015.
- [21] D. Jain, A. Lin, R. Guttman, M. Amalachandran, A. Zeng, L. Findlater, J. Froehlich. Exploring sound awareness in the home for people who are deaf or hard of hearing. *CHI*, 2019.
- [22] X. Juanola, G. Haro, M. Fuentes. A critical assessment of visual sound source localization models including negative audio. *ICASSP*, 2025. arXiv:2410.01020.
- [23] Q. Kong, Y. Cao, T. Iqbal, Y. Wang, W. Wang, M. D. Plumbley. PANNs: Large-scale pretrained audio neural networks for audio pattern recognition. *IEEE/ACM TASLP* 28:2880–2894, 2020.
- [24] H. Liu, C. Li, Q. Wu, Y. J. Lee. Visual instruction tuning. *NeurIPS*, 2023. arXiv:2304.08485.
- [25] L. May, K. Ohshiro, K. Dang, S. Sridhar, J. Pai, M. Fuentes, S. Lee, M. Cartwright. Unspoken sound: Identifying trends in non-speech audio captioning on YouTube. *CHI*, 2024. doi:10.1145/3613904.3642162.
- [26] L. May, M. Clemens, K. Dang, K. Ohshiro, S. Sridhar, P. Wee, M. Fuentes, S. Lee, M. Cartwright. “Choices? That’s the dream”: Challenges and opportunities in non-speech information closed-captioning. *Frontiers in Computer Science* 7, 2025.
- [27] A. Mesaros, T. Heittola, T. Virtanen. Metrics for polyphonic sound event detection. *Applied Sciences* 6(6):162, 2016.
- [28] C. de Lacerda Patata, M. Watkins, R. Peiris, S. Lee, M. Huenerfauth. Visualization of speech prosody and emotion in captions: Accessibility for deaf and hard-of-hearing users. *CHI*, 2023. doi:10.1145/3544548.3581511.
- [29] D. Jain et al. ProtoSound: A personalized and scalable sound recognition system for deaf and hard-of-hearing users. *CHI*, 2022. arXiv:2202.11134.

- [30] P. Revuelta, T. Ortiz, M. J. Lucía, B. Ruiz, J. M. Sánchez-Pena. Limitations of standard accessible captioning of sounds and music for deaf and hard of hearing people: An EEG study. *Frontiers in Integrative Neuroscience* 14, 2020.
- [31] A. Senocak, H. Ryu, J. Kim, T.-H. Oh, H. Pfister, J. S. Chung. Sound source localization is all about cross-modal alignment. *ICCV*, 2023. arXiv:2309.10724.
- [32] D. Jain, H. Ngo, P. Patel, S. Goodman, L. Findlater, J. Froehlich. SoundWatch: Exploring smartwatch-based deep learning approaches to support sound awareness for deaf and hard of hearing users. *ASSETS*, 2020.
- [33] K. Sung-Bin, A. Senocak, H. Ha, A. Owens, T.-H. Oh. Sound to visual scene generation by audio-to-visual latent alignment. *CVPR*, 2023. arXiv:2303.17490.
- [34] S. J. Um, D. Kim, S. Lee, J. U. Kim. Object-aware sound source localization via audio-visual scene understanding. *CVPR*, 2025. arXiv:2506.18557.
- [35] Y. Wu et al. FLAM: Frame-wise language-audio modeling. *ICML*, 2025. arXiv:2505.05335.
- [36] C. Wu et al. Qwen-Image technical report. arXiv:2508.02324, 2025.
- [37] J. Xu et al. Qwen3-Omni technical report. arXiv:2509.17765, 2025.
- [38] L. Zheng et al. Judging LLM-as-a-judge with MT-Bench and Chatbot Arena. *NeurIPS Datasets and Benchmarks*, 2023. arXiv:2306.05685.
- [39] C. Zheng, H. Zhou, F. Meng, J. Zhou, M. Huang. Large language models are not robust multiple choice selectors. *ICLR*, 2024. arXiv:2309.03882.